

Global Equity Derivatives 2016 Outlook

Fragility is the new volatility

**Bank of America
Merrill Lynch**

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Low volatility is hiding rising market fragility

Despite volatility across asset classes still below normal, we are witnessing the largest number of violent dislocations in asset prices since 2008 due to record low trading liquidity, heavy investor crowding and the manipulation of risk by central banks. Investors are suffering and hedge funds have recorded their poorest performance relative to the risk they take since 2008, despite volatility only 1/4th as high. We think today's risks are better explained by our "Fragility Indicator" which is near 80% of its '08 peaks. Leveraging signals from cross-asset risk and mispriced derivatives can help successfully navigate these challenging markets.

Trend toward higher vol firmly in place as central bank put weakens

Our call for 2015 was that volatility would begin its turn higher but remain below average, which has played out across global equities. This upward trend in volatility is only becoming more firmly established as dollar liquidity is drained from markets. Combined with a more constrained Fed as it embarks on a hiking cycle, markets may be more accident prone, and we see global volatility higher in 2016.

Top macro/directional trades: correlation gap creates cheap options

The correlation derivatives imply is not correctly priced for the reality of divergent policy and strong directional trends emerging between markets. Sell correlation to create cheap outperformance options between US, Europe and Asia, DM vs. EM, large vs. small-cap equities, Europe vs. US, and Japan vs. JPY. Low premium, low risk options could yield high payouts and offer an attractive way to trade our high-conviction directional views.

Favorite alpha trades: harvesting market dislocations

Leveraging the same dislocations driving high fragility creates opportunities for identifying high-conviction options RV. We project call overwriting will outperform in 2016 and highlight a dynamic strategy for mitigating shock risk. Despite recent oil moves, US Energy vol dispersion is too cheap to ignore. Sell dividend yield in Europe on ECB QE and hedge with credit. SX5E-HSCEI replication variance spread is attractive.

Best hedges: trade tech crowding, cheap convexity, depressed fwd vol

Buy a best-of-put on high-flying US tech stocks, which is significantly underpricing the risk of a crowded unwind. Proxy hedging overlaid on S&P put spreads outperformed 93% of the time, even adding alpha to a non-hedged portfolio. Low cost, long vol hedge in VIX offers cheap convexity. Korean forward vol cheapest long-vol to own globally on structured product depression. Best China upside and downside hedges.

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This report contains back-tested results which are based on criteria applied retroactively with the benefit of hindsight & knowledge of factors that may have positively affected performance, & cannot account for all financial risks that may affect the performance of the strategy going forward. Back-testing is hypothetical in nature & reflects application of the strategy prior to its introduction. It is not actual performance & it is not intended to be indicative of future performance.

Summary: Our outlook and top trades on a page

Table 1: 2016 Outlook & Opportunities in Global Equity Derivatives

	Theme	Implications
2016 Outlook	Rising fragility; moving deeper into uncharted waters	• Despite cross-asset volatility still below normal, we are setting records in terms of violent shocks to individual assets; 'fragility' is 80% of 2008 extremes
	Trend higher in volatility on firmer footing as liquidity drains	• With US real-rates turning higher in 2013 & a typical 2-yr lag of vol, 2015 marked a turning point in the vol cycle which only strengthens as liquidity drains
	More "local" tail risks in 2016	• Violent but short-lived shocks define today's markets owing to record low trading liquidity, investor crowding & central bank's manipulation of risk
2016 Outlook	Cross-asset risk key to gauging the risk of local going global	• Monitoring differentials in cross-asset volatility for signs a given event may fade or gather more staying power is key to navigating today's markets
	Power of CB put shows the risk of its loss	• CBs extreme sensitivity to risk since 2013 has capped vol, but the Fed's exit constrains its ability to provide the put thus increasing risks of vol rising
	Gap between correl & reality offers value	• Correlation implied by derivatives at odds with reality of divergent policy & strong trends emerging between markets; value in directional outperf. trades
2016 Outlook	Monitoring structured product trends is key for Europe & Asia	• Near-record ESTX50-linked struct. product issuance likely to be resilient in 2016. Long-dated (2y+) ESTX50 vol under pressure in moderate selloffs and supported in rallies. Asian issuance should fall from all-time highs due to regulation but remain strong. -ve spot vol correlation for HSCEI should persist if spot < 10,500. Japan 2016 issuance should pick up. Korean issuance helps bid up inter-index correlation, leading to short correlation trade opportunities
	Div outlook: Favor pull-to-par div trades in Europe & China	• ESTX50 Dec17 divs are attractive given healthy balance sheets + cheap funding. Consider HSCEI Dec16 divs (technically depressed) as an upside hedge on Chinese banks. Low EU inflation & further structured product issuance to keep downside pressure on long-dated ESTX50 divs.
	Hunting for yield in Europe as QE gains momentum	• 45% of outstanding EU govt debt is -ve yielding. With a scarcity of assets offering reasonable yields, investors should search for higher-yielding assets underpricing risk. EU HY credit and equities offer attractive yields, relative to risk, and investors may enhance income from quality div names by 2.7% via 1y, >=115% strike covered calls
	Trade/Strategy	Rationale
Macro/Directional	+Jun16 105% Best-of-call on ESTX50, SPX, HSCEI vs. -HSCEI call	• Implied correlations between SX5E, SPX & HSCEI are elevated due to dealers offloading their correlation risks from the issued structured products
	SPX over RUT 1Y ATM outperformance call contingent on RUT up	• Sell this elevated correlation (~67%) to profit from likely EU/US equity market outperformance vs. China and risk less than 2% premium
	SPX 1Y ATM call contingent on EEM < 95%	• 77% discount vs. vanilla outperformance call; 92% implied correlation. BofAML base case is for Fed tightening & lower liquidity to benefit large vs. small
Macro/Directional	SPX 1Y 95% put contingent on EEM > 100%	• 87% discount vs. vanilla SPX 1Y ATM call; 70% implied correlation. BofAML base case is for DM equities to continue outperforming EM in 2016
	SX5E over SPX Jun16 105% outperformance call contingent on SPX up	• 88% discount vs. vanilla SPX 1Y 95% put. Given record short positioning in EM and negative sentiment, a counter-trend rally in EM could materialize
	Dec16 105-120% NKY call spread contingent on USDJPY<120	• With the ECB continuing QE and Fed on track to start its hiking cycle, EU equities are poised to make up for their multi-year underperformance vs the US
Alpha/Relative Value	Jun16 105-115% TPNBNK call spread	• Investors can get significant discount given high NKY/USDJPY realized correlation; attractive because we expect further QQE to be USDJPY-negative
		• Potential policy divergence would raise Japan inflation expectations, hence benefit the banks sector, particularly in the medium term
	Trade/Strategy	Rationale
Alpha/Relative Value	Leverage X-market vol dislocations for options RV	• Instances of stark dislocation between X-market vols offer entry points for cross-asset RV between leader and laggard assets
	Enhance 'traditional' S&P500 call write programs: how to combat upside tails	• Constructive outlook on S&P call overwriting predicated on modest equity returns and higher implied vol but upside tails likely to remain a key headwind
	Buy ultra-depressed Energy vol dispersion to monetize commodity risk flares while carrying flat in a calm environment	• Mitigate upside tail risks via weekly options, consider using longer-dated options than 1M and reduce risk when recent drawdown > short call breakeven
Alpha/Relative Value	Sell long-dated EU divs, long equity as a rotation trade: Buy ESTX50	• Energy dispersion is trading at strikingly depressed levels, despite our commodity strategists warning against major downside risks in the energy space
	ATMf-107%f bullish risk reversal/ sell DEDZ9 div future or ~120%f call	• Profit from commodity risk flares while carrying positively as history suggests losses in a calm environment are highly unlikely
	Cheapen long HSCEI short ESTX50 variance relative value	• ECB QE2 likely to compress SX5E div yields: buy equity fwd/sell div future to benefit from this; high down-beta of divs offers favorable risk asymmetry.
	Trade/Strategy	Rationale
Hedges	Enhanced All-Weather S&P500 hedge	• Sell div call instead of div future to collect 7% upfront premium. In an equity rally, trade unlikely to lose as equity & div call strikes represent a (high) yield
	Buy a Best-of-put on crowded FANG stocks	• ESTX50 realized vol may be relatively contained given ECB, while spikes in HSCEI realized vol are likely as China navigates a slowing economy
	Low cost, long volatility convexity hedge in VIX	• Cheapen the HSCEI / ESTX50 var spread strike by ~5 vols by replicating the long HSCEI var leg
Hedges	Best China upside & downside hedges: Upside: Selling HSCEI Dec-16	• Combine S&P 3M 40d/25d put spreads with relatively cheap 3M 25d S&P proxy hedges to benefit from historically steep S&P skew
	1x2 ATM/114% call ratio; Downside: long HSCEI put spread	• Outperformed S&P 40 delta put 93% of time since 2008, and generated excess alpha and lower drawdowns even vs. the market in 2015
	KOSPI fwd vol is the cheapest globally, buy KOSPI & NKY Jun-16/Jun-17 fwd vol agreements	• Stellar YTD performance and crowded positioning in FB, AMZN, NFLX, GOOGL ('FANG') have increased the risk of a coordinated sell-off in these stocks
Hedges		• A 1yr 90% best-of-put on FANG trades at 54% discount to similar S&P put but during the last Tech rout, the best performing FANG dropped 3.6x the S&P
		• An 'Underfunded' VIX/SPX hedging strategy, where the long VIX OTM calls are only partially funded with short SPX OTM puts, would have provided more than 2x the hedge gains of the next best performing 'popular long VIX' strategy across major risk-off events of the past 2 years for a similar cost of carry
		• Upside: Cannot rule out a China QE; HSCEI skew is at its 91 st percentile since 2013. Upside volatility will be bid if the HSCEI rallies
Hedges		• Downside: HSCEI put was the best hedge in previous crashes but HSCEI already trades below its 2008 P/B, we prefer HSCEI put spread & KOSPI2 put
		• Asian vol offers the best value given flat term structures & low carry costs. KOSPI2 Jun-16/Jun-17 forward vol is trading at its 13th %-ile since 2008, the most subdued globally. NKY forward vol is an attractive hedge against a failure of Abenomics as policy sustainability will likely be questioned after 1Q16

Source: BofA Merrill Lynch Global Research

Rising fragility; moving deeper into uncharted waters

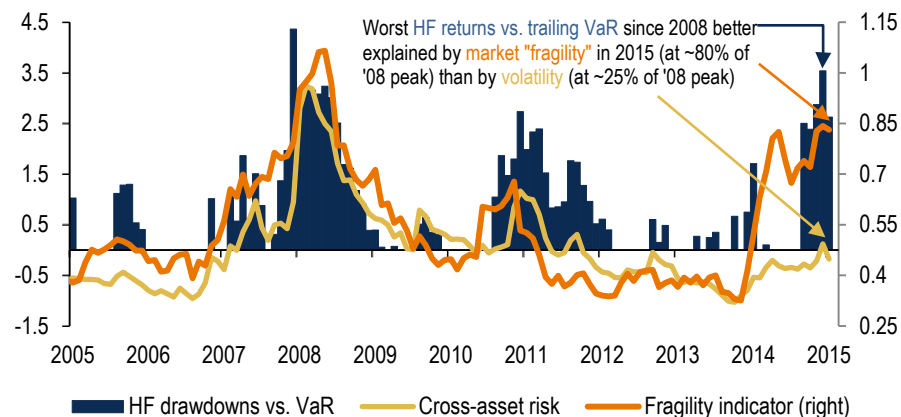
In 2016 we expect volatility to maintain its gradual upward trend, however, to continue to be punctuated with violent but short-lived shocks owing to poor liquidity, extreme positioning and a market still heavily manipulated by (and dependent on) the central bank put. Despite below-normal levels of volatility across asset classes, we are in uncharted waters in terms of a lack of stability:

- Markets are setting records in terms of jumping from calm to stressed & back
- Our indicator of cross-asset market fragility is near its highs (Chart 1)
- CB liquidity is tightening, making markets more accident prone (Chart 7)

Asset managers are struggling, with the poorest hedge fund performance relative to the risk they are taking since 2008, despite overall market volatility being only 1/4th of 2008 levels. Their poor performance is better explained by the extreme levels of market fragility, which by our metric is at 80% of its 2008 highs (Chart 1).

Unfortunately, we don't see conditions improving and only becoming more acute as liquidity continues to deteriorate, asset valuations become increasingly stretched, and the Fed navigates the unwind of the greatest policy experiment in history.

Chart 1: Hedge funds are generating their worst performance since 2008 relative to the risk they are taking, despite market vol only 25% of '08 levels. They are suffering from high & rising fragility



Source: BofA Merrill Lynch Global Research. Hedge funds are represented by an average of the HEDGNAV, HFRIFWI, and EHFI251 Indices. Performance is shown on the left axis as max drawdown / trailing 2Y 95% VaR using monthly returns. Macro funds have been particularly hard hit in 2015, recording some of their worst absolute performance since 2008. Cross-asset risk (left axis) is an average of 15 volatility and credit risk measures from GFSI, measured as z-scores. These are the same measures used in the fragility indicator defined on p. 7.

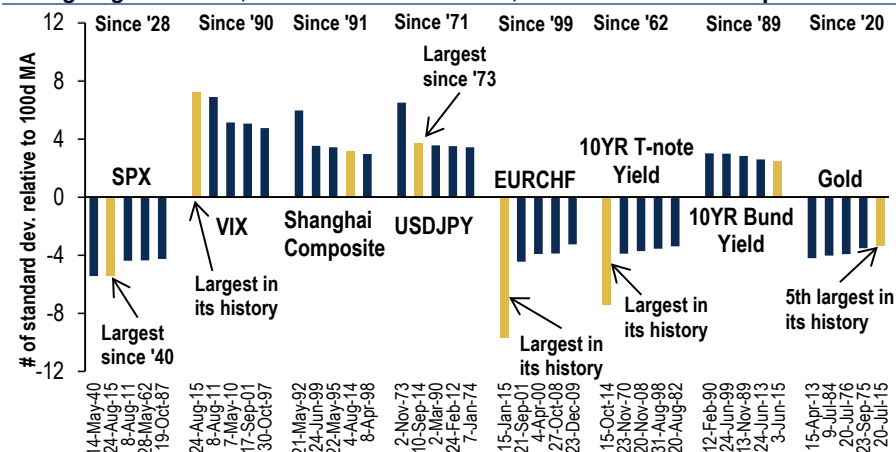
Successfully trading a less stable world

While many are struggling with this new market dynamic, we believe there are smart ways to combat it – and even profit – by monitoring cross-asset risk, and taking advantage of the inflection point in asset correlations. For example:

- Gaps in cross-asset volatility can aid in differentiating between “local” and global risks, to determine when to fade the market or add a hedge (p. 8)
- Shocks create entry points for cross-asset RV between leader and laggard assets which has been successful in generating alpha (p. 23)
- Risks implied by derivatives, including correlation, often are unlikely to realize as stress unfolds, allowing for cheap directional trades (p. 11)
- Cheaper hedges can be constructed by harvesting underpriced volatility through proxy puts overlaid on standard put spreads (p. 30)
- Strategies that collect the volatility risk premium, for example through call overwriting, while dynamically managing these risks can add alpha (p. 24)

As we highlighted in our 2015 outlook, the most distinguishing feature of markets today is not the general trend in volatility, but the unprecedented turbulence.

Chart 2: We have seen an inordinate number of assets experience record setting jumps from calm to stress since summer 2014. The S&P 500 recorded its largest 1-day jump from its 100-day trading range since 1940, 10-Yr treasuries since 1962, and EURCHF since its inception in 1999



Source: BofA Merrill Lynch Global Research. Daily data of the SPX from 30-Dec-27 to 2-Dec-15, VIX from 2-Jan-90 to 2-Dec-15, SHCOMP from 31-Dec-90 to 2-Dec-15, USDJPY from 31-Mar-71 to 2-Dec-15, EURCHF from 31-Mar-99 to 2-Dec-15, USGG10YR from 30-Mar-62 to 2-Dec-15, GDBR10 from 31-Mar-89 to 2-Dec-15 and XAU from 31-Mar-20 to 2-Dec-15. We measure this one day dislocation as the magnitude of the level relative to its trailing 100d moving average in standard deviations. If we see records set on subsequent days, we only record the largest as representing that event. For USGG10YR, we use the intraday move on 15-Oct-14.

Vol rises off lows; equity vol still muted

Our 2015 year-ahead call was that equity volatility would rise from its 2014 lows but more slowly than many expected and would average below long-term norms.

So far this year, the VIX (US), V2X (Europe), VNKY (Japan), and VHSI (Hong Kong) are all above 2014 levels but below long-run averages (though some are close, Chart 4).

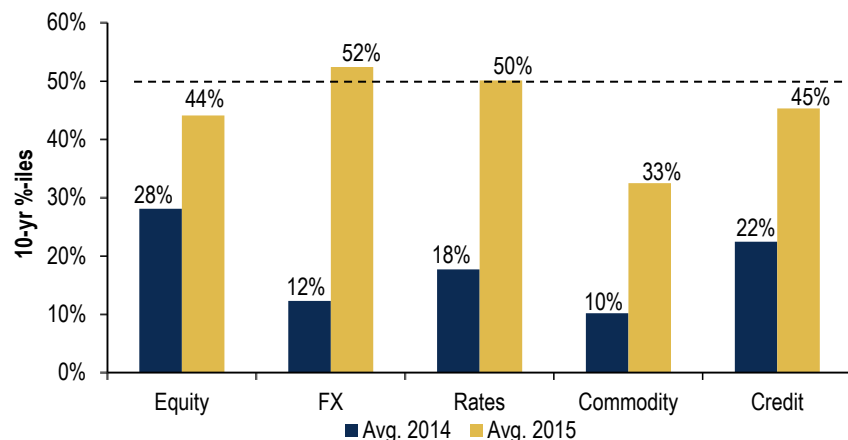
US equities also realized less volatility than the average since 1928, however, volatility was in-line with the average level during years when S&P has risen (Chart 5).

In terms of cross-asset volatility, in 2015 we witnessed a divergence between equity vol and credit spreads, which have remained below long-term norms, and rates and FX vol which rose faster in anticipation of a global policy divergence (Chart 3).

In 2016, we expect to see more convergence in cross asset risks as the largest impact of policy divergence on rates and FX may be behind us. Our US rates team [anticipates volatility to decline](#), once rate hikes begin, as the certainty of a steady policy path is lower than the uncertainty of a Fed on hold.

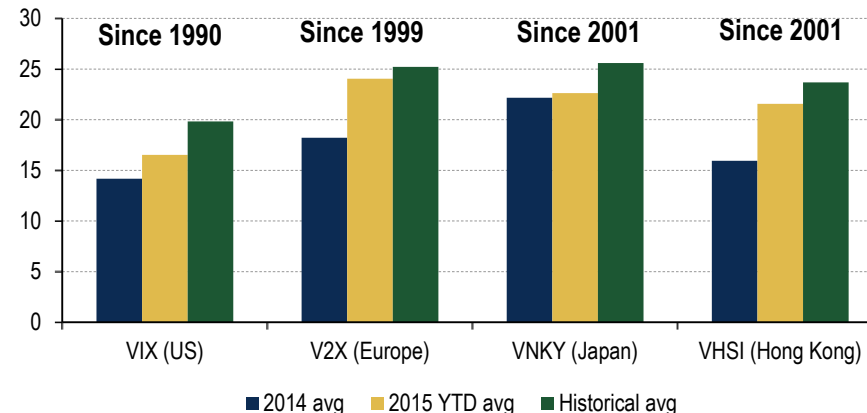
We also believe equity volatility will continue its upward trend as liquidity continues to drain, and given the Fed may have a more difficult time reacting to market weakness for fear of being seen as committing a policy mistake.

Chart 3: Only FX and Rates volatility averaged above their 10-year medians in 2015, equity and commodity vol, and credit spreads were lower than normal. In 2016 we expect to see more convergence in cross asset risks



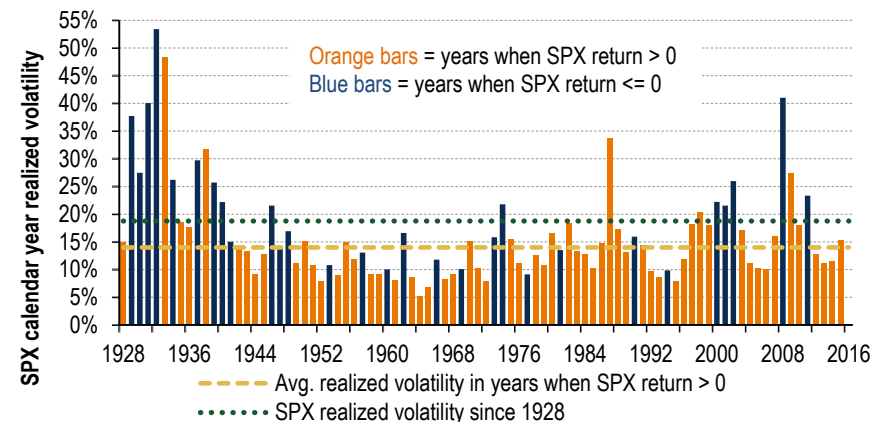
Source: BofA Merrill Lynch Global Research. Daily data from: 1-Dec-05 for Equity (VIX Index), Rates (MOVE Index), FX (avg. 3m implied vol on EUR/USD, USD/JPY, USD/CHF), Credit (US High Grade Corporate, US HY Corporate), 1-May-07 for Commodity (Merrill Lynch Commodity 3 month volatility Index). All data through 1-Dec-15.

Chart 4: In line with our forecasts, equity implied volatility across the regions rose off their 2014 lows this year, however, remained below long-run averages



Source: BofA Merrill Lynch Global Research. Daily data of the VIX from 2-Jan-90 to 1-Dec-15, V2X from 4-Jan-99 to 1-Dec-15, VNKY from 4-Jan-01 to 1-Dec-15 and VHSI from 2-Jan-01 to 1-Dec-15.

Chart 5: The S&P 500 YTD has generated below long-run average levels of volatility, however, is close to the average volatility for years when the S&P has risen since 1928



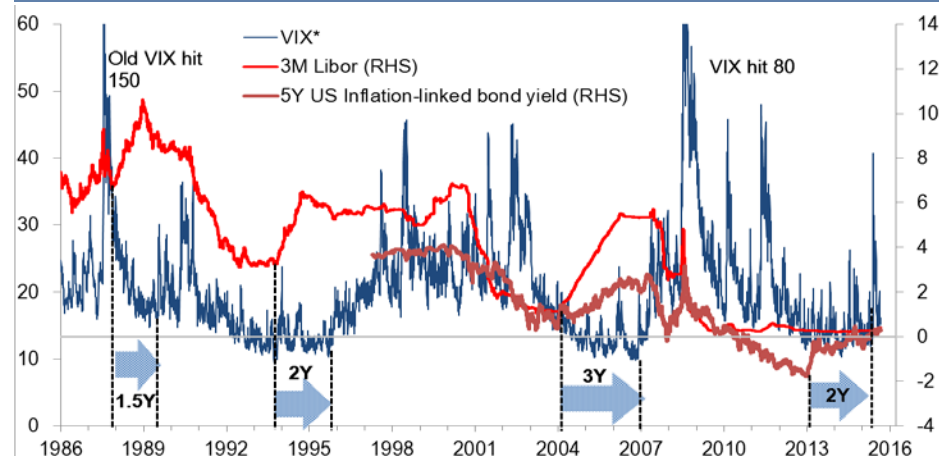
Source: BofA Merrill Lynch Global Research. Annual data from 1928 through 2015 (as of 2-Dec-15).

A trend of rising vol as liquidity drains

Our Economics of Volatility¹ framework has been anticipating the 2015 starting point to a turn in volatility for the last two years². From here on we expect to see a rising trend in equity volatility levels, a trend that could last 1-2 years, transporting us from the low volatility regime of the last 3 years towards a sustained high volatility regime.

Our expectation for a turn in the volatility cycle follows from a clear turn higher in 5Yr real rates in 2013, and allows for a 2-year lag (Chart 6). High volatility regimes may resemble periods like 1998-2003 or 2008-2011, as two examples. Transition periods can also take various forms. Unlike the 1996-1998 transition period which was gradual and well behaved, the 2008-2009 transition was short and violent, as a suppressed and overdue re-pricing of risk finally manifested itself. It's hard to predict the exact form the next transition will take. While our base case is for an orderly transition, we are wary of the possibility of unpleasant surprises resulting from an unwinding of the highly unusual monetary policy of the last 7 years.

Chart 6: The relationship between the interest rate cycle and equity volatility: 2-year lead. To cope with ZIRP, we use the 5Yr TIPs rate as an alternative measure of monetary policy in place of LIBOR



Source: BofA Merrill Lynch Global Research. Data from Mar-86 to Dec-15

Unwinding extreme easy monetary policy is a tightening

The monetary tightening cycle which started with the 2013 taper has continued its progress, reflected in rising 5yr real rates. This in turn has driven a significant tightening in global liquidity as capital flows from developed to emerging markets start to reverse, evidenced in a slowdown and reversal of FX reserve accumulation.

¹ See "The Economics of Volatility – It's a Liquidity Thing", published 31 Jan 2005

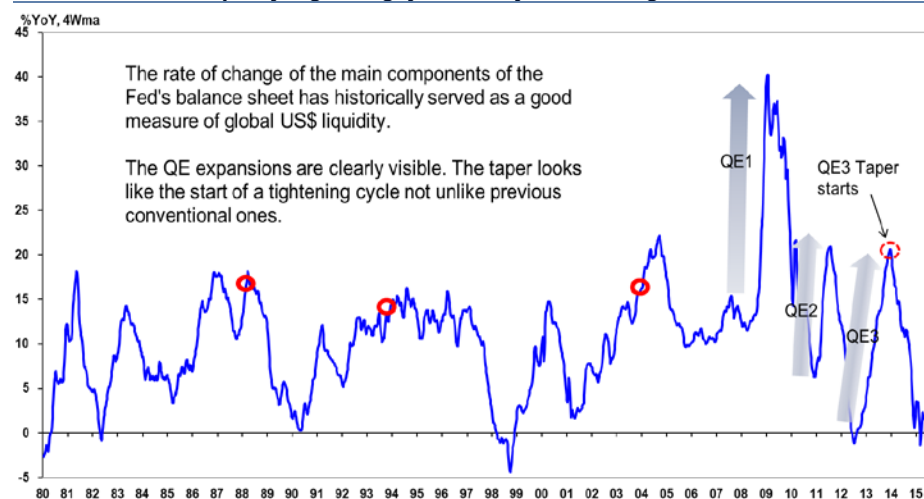
² See "Global Equity Derivatives 2014 Outlook", published 10 Dec 2013, and "Global Equity Derivatives 2015 Outlook", published 9 Dec 2014

Tightening liquidity combined with fragility: equity markets are accident prone

Chart 7 shows a measure of global US\$ liquidity derived from the momentum of the Fed's balance sheet. Historically we see that tightening cycles have typically started at high liquidity levels. The current cycle in fact started in anticipation of the tapering of the open-ended QE3 program in 2013, with the impact evident in the sharp turn in the 5Yr TIPs rate (Chart 6), and the subsequent fall in US\$ liquidity.

Given how far liquidity has already dropped, it is going to be interesting to watch the impact of the more traditional part of the tightening cycle – actual rate hikes – which are expected to start imminently. Combined with our view of an increased likelihood of local shocks due to deteriorating trading liquidity, we may find the markets more accident prone in 2016 than they have been in some time.

Chart 7: Global US\$ Liquidity: Tightening cycles usually start near highs (circles denote starts)



Source: BofA Merrill Lynch Global Research. Data from Jan-80 to Dec-15.

Composition: US Monetary Base (BBG: ARDINTBA & ARDIMONY) + Treasury & Agency securities held in custody by the Fed for foreign official and international accounts (BBG: FARBTRE, FARBAGSE)

The Economics of Volatility framework – a quick refresher

The framework attempts to forecast the transitions from low volatility environments to high volatility ones, and vice versa. These periods can last a few years at a time, as do the transitions themselves. The ends of peak and trough periods in interest rates lead declines and rises in mean volatility levels by around 2 years. During low volatility environments, the inevitable volatility spikes that occur as new information hits the market will get washed over with plentiful risk capital, such that the spike is quickly suppressed. In high volatility environments, higher mean volatility levels can be sustained due to a reduced supply of risk capital.

Rise of the “local” tail risks

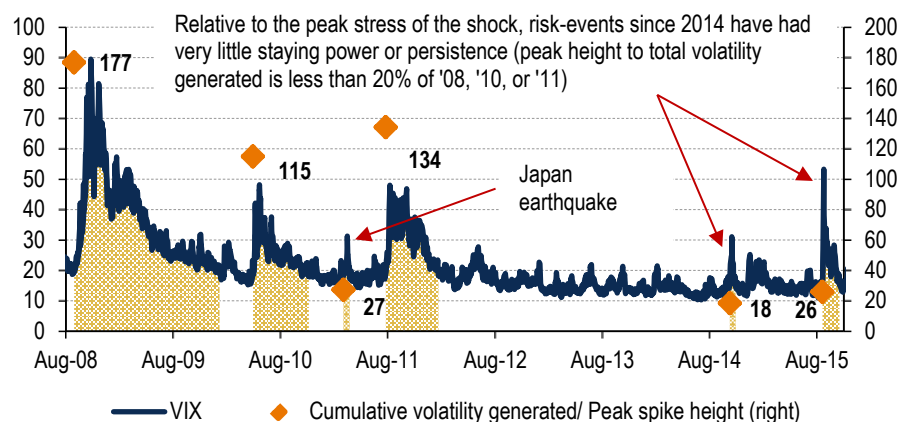
The key theme introduced in our 2015 year ahead, [Storms in a teacup as vol finally starts its turn](#), was that we would see more “local” shocks or violent moves that were relatively short-lived, as trading liquidity (bank balance sheets and fickle high frequency capital) continue to create a toxic environment prone to tantrums and flash-crashes.

This indeed continued in 2015, and US equity markets set 75-year records in August in terms of their ability to jump from states of calm to states of stress (Chart 8). Despite the VIX exceeding 50 intraday (only surpassed in 2008), it collapsed at near record speed with the average VIX level still below long-term norms.

We expect this trend to continue in 2016, as trading liquidity is only getting worse, and post-08 policy aimed at reducing extreme tail risks is actually accentuating local risks.

The crucial question of whether these “technical” shocks fade fast is a function of whether central banks maintain their credibility as a crisis fighter of last resort. Our base case is that they do, however, we remain in uncharted territory and with only downside risk to credibility.

Chart 8: In traditional market shocks such as 2008, 2010, and 2011, the ratio of the “total volatility generated” by the event to its peak height was consistently over 100, however, local shocks have recorded high volatility with little staying power and ratios less than 30



Source: BofA Merrill Lynch Global Research. Daily data from 1-Aug-08 through 3-Dec-15. Cumulative volatility generated = sum of VIX levels during a volatility spike (corresponding to shaded areas in the chart).

Measuring fragility

Normally, there is a strong relationship between the size of a market shock, and the amount of volatility generated by the aftershocks before markets calm to normal levels. However, in a local tail event the relationship is very different with a tendency for markets to see extreme dislocations which reverse at record pace. This is precisely what we have seen over the last 18 months across asset classes (Chart 2).

Illustrating this for US equity volatility specifically, Chart 8 shows the relationship between the peak height in the VIX to the total volatility generated during the shock before it retraced back to pre-spike levels. It shows that since 2014, we have been in uncharted territory in terms of large shocks occurring with very little total damage done to markets. This allows average volatility to remain muted but creates challenging conditions for managing risk and generating returns.

The Fragility Indicator

Extending this concept of sharp dislocations to cross-asset volatility, our Fragility Indicator measures the frequency and extent to which volatilities and credit risks in our GFSI index are witnessing abnormally large shocks, or outlier events from the group.³ So for example on 24-Aug-15, S&P 500 implied volatility rose to extremes versus other risks in GFSI, as shown in Chart 9, which indicated high fragility.

Normally, the magnitude of this fragility indicator is directly proportional to the total level of volatility. However, we have recently entered uncharted territory in terms of dislocations rising to near 2008 levels, despite average volatility still below normal, as shown in Chart 1.

As shown in Chart 2 earlier, the market across asset classes has in the last 18 months generated a large number of extreme outlier events that in numerous cases have set multi-decade records. It is these events that continue to drive our fragility indicator to such high levels. However, as these episodes have not resulted in significant risk-contagion (in part a result of the power of the central bank put), overall volatility levels have remained muted. In other words, we have recorded a significant number of local shocks that have not become global.

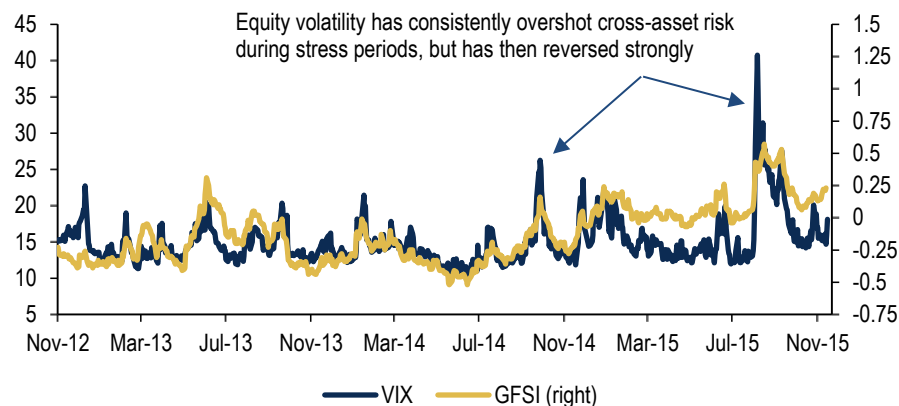
³ The Fragility Indicator is based on 15 GFSI sub-components (z-scores) spanning global equity vol (5 factors), credit spreads (4), FX vol (3), rates vol (2), and broad commodity vol (1). Specifically, we calculate the cross-sectional dispersion of each risk factor vs. its peers [i.e., (risk factor level – average of others) / stdev of all] and record the largest 5-day change in dispersion across the 15, conditional on the underlying risk factor rising. We then average these readings over a rolling 6M window.

Gauging the risk of local going global

One of the key reasons we believe local shocks occur is that they are driven by technicals including crowded positioning and markets' malfunctioning from a lack of trading liquidity. However, with little fundamental depth, they fade quickly, aided (in particular) since 2013 by central banks that have never been more sensitive to market risk.

A key sign of a technical shock is one in which risk is dominated by a single asset class, without a fundamental re-pricing of risk across markets. This was precisely what we witnessed in both October 2014 and August 2015, when equities recorded as much as five times the stress of any other asset class (Chart 9). Monitoring this gap is key to deciding whether to fade a shock or add hedges for a more prolonged event, as major risk-off events are never isolated to a single asset class. It also creates opportunities for cross-asset RV as discussed on [p. 23](#).

Chart 9: When a single asset class vol severely dislocates from broader risk measures, it doesn't last, as either contagion sets in or risk reverses. This disconnect can be used to gauge the fundamental depth of a shock, and creates a good opportunity for relative value trades

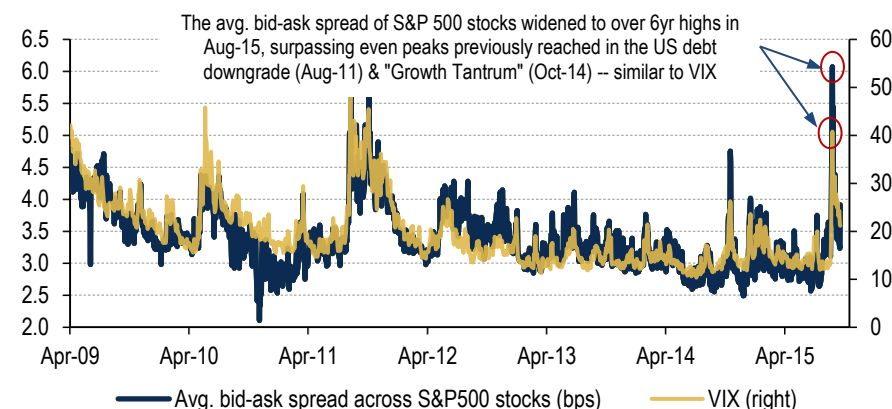


Source: BofA Merrill Lynch Global Research. Daily data from 19-Nov-12 through 2-Dec-15.

Houston, we have a liquidity problem

Falling market liquidity has been a key driver of these local shocks. While regulations constraining bank balance sheet-intensive fixed income markets are well discussed, what is less appreciated in our view is that liquidity in agency-based electronic markets is at record lows. Not only is high frequency capital that provides the majority of liquidity in these markets becoming more risk-averse during times of rising stress, but regulation created to prevent events such as the May-10 flash crash are exacerbating the risks of an Aug-15 style event.

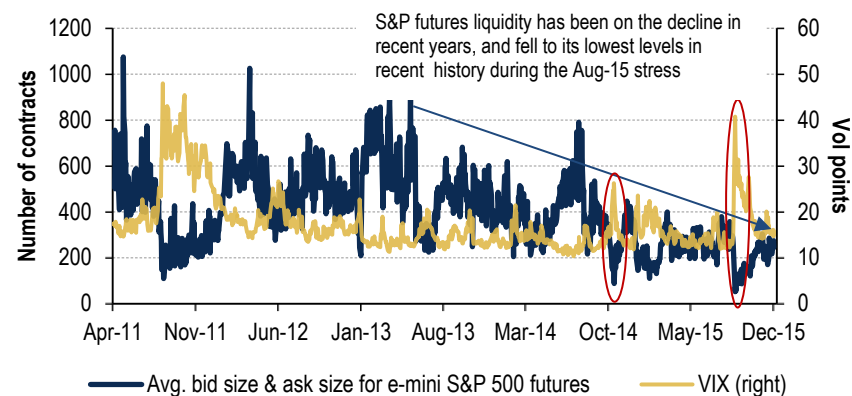
Chart 10: S&P 500 bid/offer spreads hit extremes not recorded since 2008 in Aug-15, at the same time the VIX hit its highs since 2008 and the S&P recorded its largest 1-day dislocation since 1940



Source: BofA Merrill Lynch Global Research. Based on tick data from 1-Apr-09 through 17-Sep-15.

Single stock and futures trading halts, combined with the uncertainty of not knowing at what levels of stress trades may be canceled, creates significant risks for liquidity providers. As a result, in the Aug-15 sell-off, average bid-offer spreads of all stocks in the S&P 500 intraday, reached their highest level since at least 2008 (Chart 10). At the same time, the top-level order book depth of S&P 500 e-mini futures hit its lowest levels since at least 2011 (Chart 11). Both occurred as the VIX was hitting its highs, illustrating how sensitive liquidity providers are to risk.

Chart 11: The depth of the S&P 500 e-mini futures market also hit historical lows in Aug-15

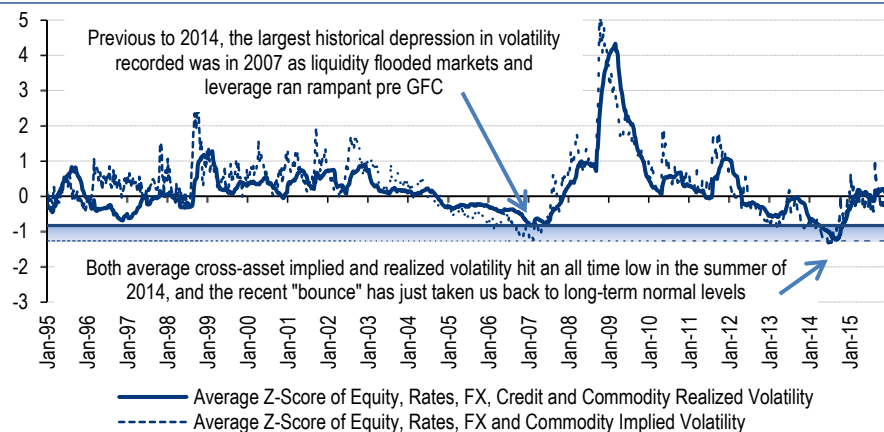


Source: BofA Merrill Lynch Global Research. Based on Level I order book tick data from 1-Apr-11 through 2-Dec-15.

Power of CB put shows risk of its loss

Central banks have had a tremendous impact on financial markets in the last seven years, which is never more apparent than when looking at the world through the volatility lens. As shown in Chart 12, cross-asset volatility reached all-time lows in the summer of 2014, falling even below the 2007 pre-GFC bubble lows, crushed under the weight of unprecedented monetary policy (or in the ECB case, the promise of policy). This is remarkable considering the size of the risk “bubble” created pre-GFC.

Chart 12: Cross-asset volatility fell to its lowest levels in history in the summer of 2014, even below 2007 “bubble” lows, illustrating the massive impact that experimental monetary policy has had on risk assets



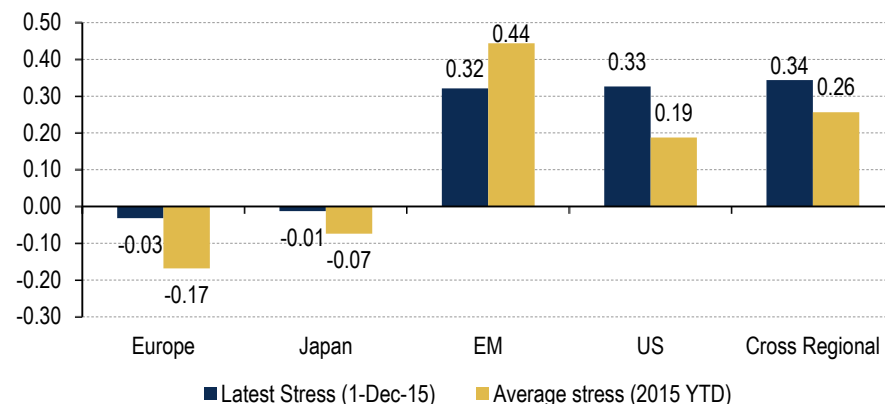
Source: BofA Merrill Lynch Global Research. Implied volatility data from: 5-Jan-90 for Equity (VIX Index) and Rates (MOVE Index); 1-Jan-99 for FX (avg. 3m implied vol on EUR/USD, USD/JPY, USD/CHF); 4-May-07 for Commodity (Merrill Lynch Commodity 3 month volatility Index). 6M realized volatility data from: 3-Jan-95 for FX (EUR/USD, USD/JPY), Equity (S&P500) and Rates (10 Year US Swap Rate); 1-Jul-97 for Credit (US High Grade Corporate, US HY Corporate); 6-Jul-00 for Commodity (D) UBS). All data through 30-Nov-15.

The result is that risk is not fairly priced based on fundamentals but rather is better explained by investors not wanting to stand in front of central banks as they embark on QE. As Chart 13 shows, when decomposing the 41 factors of risk covering 5 asset classes from our [GFSI index](#) into regions, both Europe and Japan (the two regions still actively engaging in QE) are the two regions with the most depressed price of risk. This is despite being the two developed regions with some of the greatest fundamental risk.

Unprecedented CB – market co-dependence

Central banks have never been more sensitive to financial market conditions as they are today. This hyper-sensitive reaction function has placed huge downward pressure on volatility, and has accentuated local shock behavior as investors have become accustomed to CBs verbally supporting the market at very low levels of stress compared to the past.

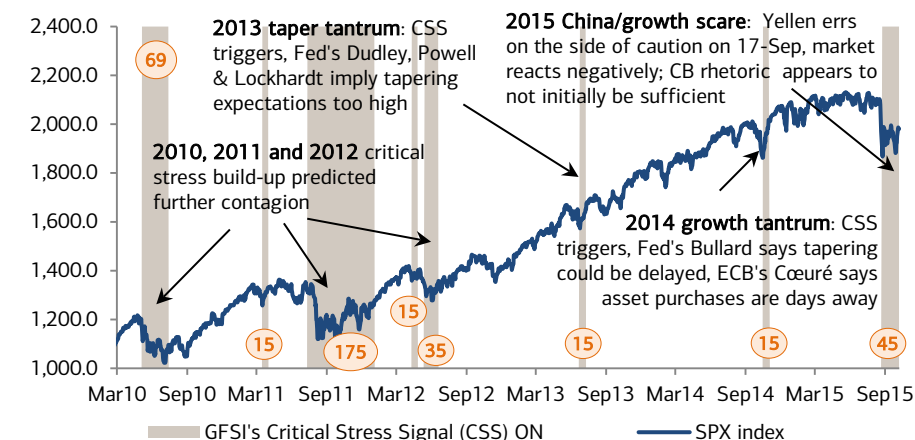
Chart 13: Europe and Japan remain the two markets with the most depressed price of risk, despite being among the least fundamentally sound, as no one has wanted to stand in front of the ECB or BOJ as they embark on QE



Source: BofA Merrill Lynch Global Research, BofA Merrill Lynch Global Financial Stress Index

In the last three instances when our GFSI [critical stress signal](#) has triggered, during the taper tantrum in June 2013, the Oct 2014 growth tantrum, and the Aug 2015 China tantrum, central banks have stepped in to verbally support the market (Chart 14).

Chart 14: From 2000 -2012, 12 out of 15 times the GFSI critical stress signal triggered, stress continued to rise and equities fell. Then beginning in 2013 with the taper tantrum, in every instance the critical signal triggered, central banks have verbally reversed market stress



Source: BofA Merrill Lynch Global Research, BofA Merrill Lynch Global Financial Stress Index. Circled numbers are number of days the Critical Stress Signal was on.

In each case central banks have reversed market stress, creating a string of three false signals, which is historically unusual. From 2000-2012, the GFSI's critical stress signal triggered 15 times, 12 of which resulted in a further escalation of risk and a pull-back in global equities of at least 5%. This illustrates the extent to which central banks have essentially capped risk at levels where it historically was likely to spill over.

This has self-reinforced a “buy-the-dip” mentality which, together with the fact that investors have generally been underweight US equities this year, has caused the S&P to record larger returns on days the market was rising than when it was falling. Combined with the fact the S&P fell more days than it rose YTD but the market overall was up makes this historically unusual, occurring only 5 other years since 1928.

Pulling the safety net away will be risky

Arguably one of the reasons central banks have been so sensitive to market risk is that they are fearful of a negative wealth effect resulting from a financial market sell-off hurting the real-economy, given they have little monetary ammunition left. Keeping rates low to avoid the rising costs of record high debt burdens could also be a motive.

The US Fed's fear was made particularly clear by Yellen's decision to not hike in September, citing the sell-off in equities and China weakness, at a time when the S&P 500 was only about 10% below all-time highs.

However, the challenge will be to remove this safety net given how dependent the market has become. And once the Fed begins its hiking cycle, it may be implicitly less able to provide the support for fear of being seen as making a policy mistake. This reduced power of the CB put will only help increase market fragility.

Central bank's risk manipulation well explains local tails

A good way to explain why we have seen local tail risks arise so frequently since central banks began to heavily manipulate asset prices is with the following analogy, illustrated in Exhibit 1. Essentially central banks, by unfairly inflating asset prices have compressed risk like a spring to unfairly tight levels. Unfortunately, the market is aware the price of risk is not correct, but they can't fight it, and everyone is forced to crowd into the same trade. By manipulating markets they have also reduced investors' inherent conviction by rendering fundamentals less relevant.

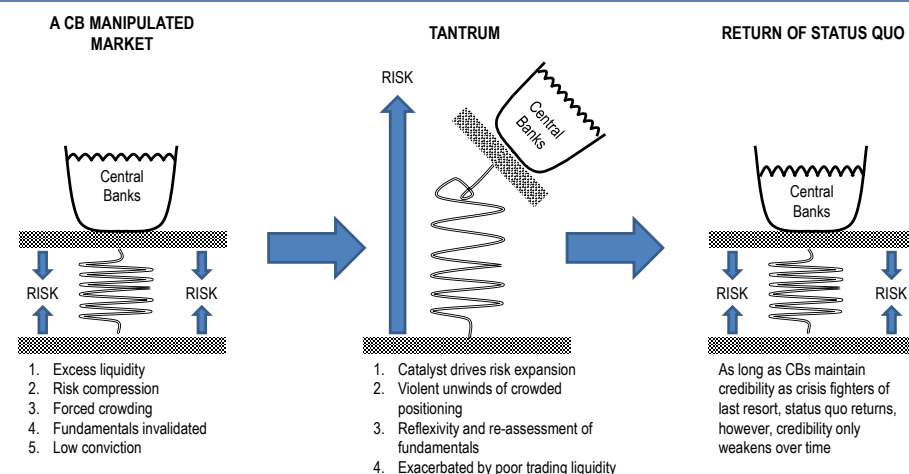
This then creates a highly unstable (fragile) situation that breaks violently when a sufficient catalyst causes risk to rise – overly crowded positioning meets a market with little conviction.

Catalysts can range from a “valuation scare” similar to Oct-14 or Aug-15 to a prominent investor stating that assets (e.g. bunds) are not fairly priced and are the “short of the century”.

The unwinds from these crowded positions are violent, but almost equally violent in some cases are the reversals, which are driven from investors crowding back in when they realize central banks are still there providing protection.

From this vantage point, it becomes clear that the biggest visible risk to financial markets is a loss of confidence in this omnipotent CB put.

Exhibit 1: Evolution of Central Banks' induced local shock



Source: BofA Merrill Lynch Global Research

Gap between correl & reality offers value

Historical correlation tends to influence the pricing of implied correlation in derivatives markets.⁴ However, 2016 could mark a critical inflection point in many cross-market and cross-asset correlations, owing to the potential effects of policy divergence and shifts in positioning or sentiment. We see value in monetizing probable correlation disconnects for either cheap macro directional trades or efficient hedges. Importantly, even if correlation is priced “correctly”, directional option holders can benefit from divergent price paths. For example, EEM has under-performed the S&P 500 by 16% over the past year, yet the 1Y realized correlation of weekly returns is 75%. The latter leads to high implied correlation that can be sold to cheapen EM vs. DM trades.

Policy divergence: Likely a game-changer for correlations

Our economists expect [policy divergence](#) to be a key theme in 2016, as the ECB, BOJ, and PBOC all ease further while the Fed normalizes policy. In our view, this could create gaps between current and future asset correlations. For example:

- Average pairwise realized correlation between the S&P 500, ESTX50, and HSCEI (68%) is not far from 5yr highs, in contrast with our divergent outlooks for the US, Europe, and China ([p. 19](#)).
- Realized correlation between the S&P 500 and EEM (75%) has rebounded sharply since summer 2014, however, BofAML’s base case is for DM equities to continue outperforming EM in 2016. Alternatively, given record short positioning in EM and negative sentiment, a counter-trend rally in EM equities could also materialize, driving EM > DM divergence. Correlation can be harvested either way ([p. 20](#)).
- Realized correlation between the NKY and USDJPY (73%) is within striking distance of YTD highs, however, our strategists expect potential further easing by the BOJ in 2016 to be equity-positive yet USDJPY-negative ([p. 21](#)).
- Realized correlation between the ESTX50 and S&P 500 (70%) is near YTD highs, however, our equity strategists anticipate greater upside for European equities in 2016, in part due to divergent ECB and Fed policies ([p. 20](#)).
- The S&P 500 and Russell 2000 remain highly correlated (88%), but our US equity strategists expect Fed tightening & lower liquidity to benefit large vs. small ([p. 20](#)).

Sell elevated correl for cheap macro directional trades

We highlight our top macro directional ideas related to the above policy divergence themes on pp. [18-21](#). These are broadly “outperformance” trades, 6M to 1Y in tenor, that leverage elevated implied correlation to generate discounts of up to 70%-90% vs.

⁴ Note that realized correlation is not the only driver of implied correlation. Retail structured product issuance as well as supply/demand in the broker-dealer market for correlation are also relevant factors.

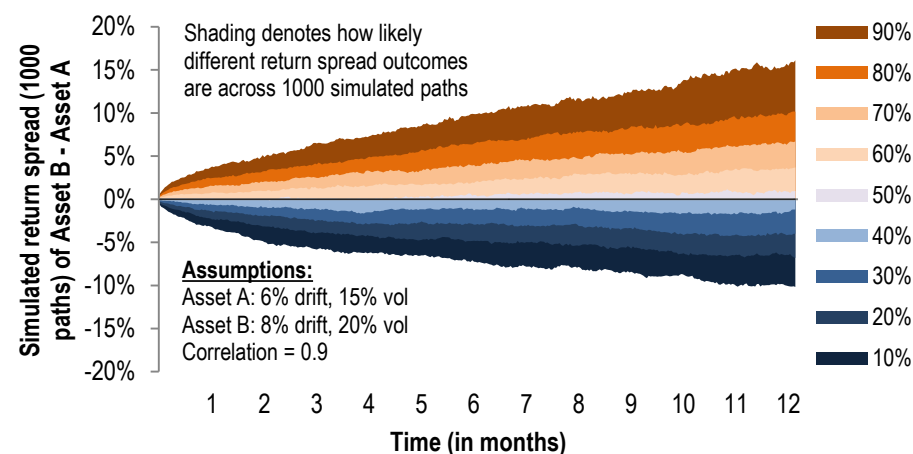
more vanilla implementations. Risk is limited to the upfront premium, which is often sub-1%. This in turn creates potential for high payout ratios, particularly considering that neither far out-of-the-money strikes nor tight option spreads are involved.

An important distinction should be drawn between the realized correlation generated by two assets during the life of an outperformance-type trade and their terminal return spread, where relative beta is important. The former matters more to the dealer risk-managing the position while the latter is more relevant to the end-user. Chart 15 shows that even well-correlated assets (90%) can generate price paths that diverge meaningfully over the course of 1yr, benefiting the end-user. In our view, this is analogous to the value a directional holder of a vanilla option (who cares more about asset return) can extract vs. a delta-hedger (who cares more about asset volatility).

Buy correl via cheap hedges for underpriced crowding risk

Implied correlation can also be an attractive buy, particularly in corners of the market where derivatives appear to be underpricing the risk of a coordinated sell-off driven by a reversal in crowded positioning, sentiment, or market leadership trends. We believe this to be the case with the red-hot “FANG” stocks (Facebook, Amazon, Netflix, and Google), which have added \$635bn to S&P market cap in 2015 while the remaining index constituents have lost \$434bn, thus coercing many investors into “long FANG” exposure for alpha. With 1Y implied correlation of these stocks ~50%, long correlation hedges can be constructed that could be multiple times more efficient than S&P-based protection should crowded FANG positioning unwind in 2016 (see [p. 31](#) for details).

Chart 15: Even two well-correlated assets can generate price paths that diverge meaningfully over time. Herein lies the value for directional users of outperformance-type options



Source: BofA Merrill Lynch Global Research. Paths were generated assuming asset prices follow correlated geometric Brownian motions.

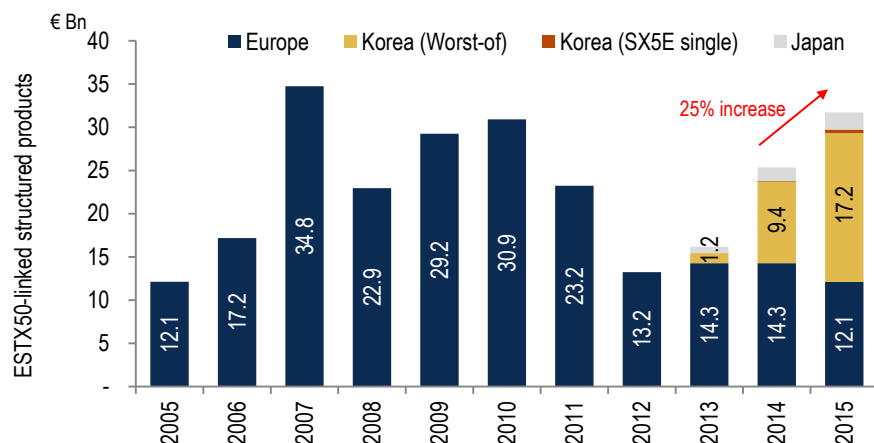
Monitoring structured product trends is key for Europe & Asia

We have previously highlighted how the issuance of equity index-linked retail structured products drives several phenomena, including the [Asianification of volatility](#), downward pressure on [long-dated dividend futures](#), elevated [DM-EM equity index correlation](#) and dislocated [repo \(borrow\)](#) curves. Thus, monitoring product issuance trends and dealers' outstanding exposure can help to identify trading opportunities in multiple markets. In 2016, we believe this will remain an important theme even if Asian issuance (which has led rapid product growth in recent years, Chart 16) slows. Indeed, Asian retail investors are increasingly looking beyond their region for equity exposure and the ECB's QE programme has given them good reason to look to European assets.

Near record ESTX50-linked issuance likely to be resilient

2015 is on track to be the second biggest issuance year (after 2007!) in 10yrs for ESTX50-linked structured products globally. The strong (25% YoY) growth of (largely income-oriented) ESTX50-linked structured products has been driven by Korea, as Chart 16 illustrates, with the majority of products being auto-callables. Despite the risk of a decline in overall Korean issuance (see p. 13), ESTX50-linked issuance is likely to remain supported due to (i) regulatory constraints on HSCEI-linked issuance that could increase the ESTX50's market share, and (ii) typically higher demand for auto-callables on indices that appear to have greater upside potential (which given ECB QE, EU equities seem to offer in 2016).

Chart 16: The growth in issuance of (income-oriented) ESTX50-linked structured products, mainly out of Korea, has ramped up in recent years (the majority being worst-of auto-callables)



Source: BofA Merrill Lynch Global Research. Data: 1-Jan-05 to 6-Nov-15 (2015 issuance is annualized as of 6-Nov-15).

Dealer hedging of ESTX50 structured products has significant market impact

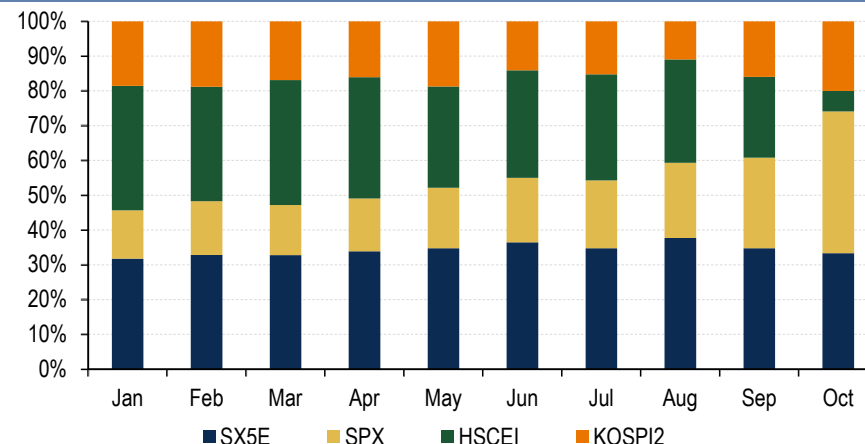
ESTX50-linked structured product issuance in 2015 (annualized) is likely to be €32bn, a 25% YoY rise, equivalent to ~€140mn of ESTX50 vega. We estimate the current outstanding vega of ESTX50-linked structured products to be ~€100mn (Chart 18). To put this into context, the open interest notional of all outstanding listed ESTX50 options with 2-5y maturity is ~€220bn (€1bn vega). While this suggests only a 10% vega contribution from ESTX50 structured products to long-dated listed options, dealer hedging of products is likely to have a **significantly greater impact**, as:

- Synthetics/box spreads (which have 0 vega) and spread option structures (double-counting of vega) lead to significant overestimation of ESTX50 listed option vega
- Dealer hedging of structured products is biased the same way and is more pronounced in sharp market movements (when there is likely little other long-dated option flow) vs more offsetting (buy/sell) flows in the broader listed market
- Dealer hedging is likely to be more concentrated in long-dated puts whereas we include both calls & puts in our broad listed option vega estimate.

Vega peak of outstanding ESTX50-linked structured products at ~3150 level

According to our estimates, the current peak of issuers' vega exposure from outstanding ESTX50-linked structured products is ~3150 (i.e. 10% below the ESTX50 spot on 30-Nov, Chart 18). Notably, the outstanding vega profile from Korean issuance differs significantly vs that of European issuance (Chart 18), primarily due to the

Chart 17: SX5E and SPX have become more popular underlyings for auto-callables issued out of South Korea



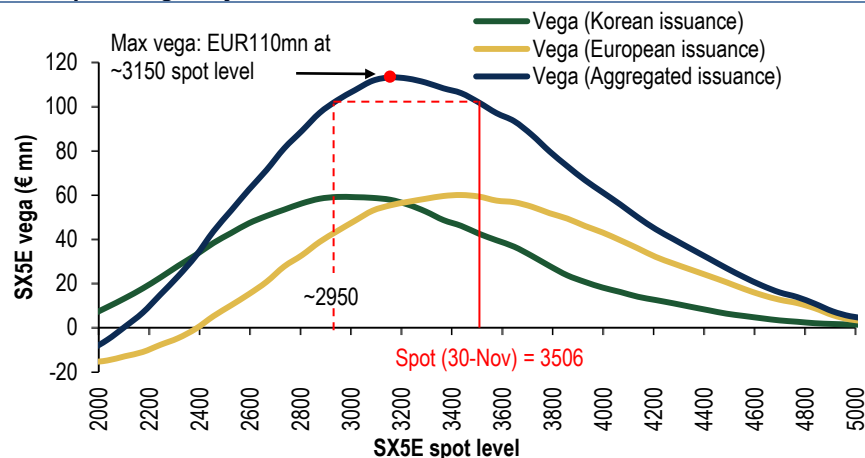
Source: BofA Merrill Lynch Global Research. Data is from 1-Jan-15 to 31-Oct-15.

difference in the nature of issued products in the regions. While a typical auto-callable issued in Europe has a maturity of 5-6y and has a fixed upside knockout barrier (usually ATM, ie where the product is called away), products issued out of Korea are of ~3y maturity and tend to have step-down knockout barriers.

HSCEI movements could materially impact ESTX50 vega profile: With Korea dominating ESTX50-linked structured products issuance, the outstanding vega profile is affected by the behavior of the HSCEI, SPX & KOSPI. This is because a typical auto-callable issued out of Korea sells a worst-of put on a basket of indices (eg ESTX50, SPX, HSCEI & KOSPI). As a result, a move in either of these underlyings affects the partial vega of the other. For example, as shown in Chart 19, ESTX50 vega outstanding for the same levels of the ESTX50 is materially different for +/- 20% moves in the HSCEI.

Long-dated ESTX50 vol likely to be under pressure in moderate selloffs & supported in rallies: Long-dated ESTX50 vols are likely to underperform in moderate ESTX50 sell offs and only pick up materially with spot <2950 (dealer vega at 2950 = vega as of 30-Nov, Chart 18). In case of a market rally, particularly a sharp one, long dated vols are likely to be supported unless new issuance is significantly greater than called-away product notional. **Trade: Investors could therefore consider selling ESTX50 long dated (Dec17+) call ratios** to benefit from this phenomenon, with a view to unwind it in 6m-1y time such that the structure retains high vega (& low gamma) sensitivities.

Chart 18: Current peak of issuers' ESTX50 vega exposure from outstanding ESTX50-linked structured products globally is ~3150



Source: BofA Merrill Lynch Global Research. Data is as of 30-Nov-15

Asian issuance set to decline in 2016 but still important

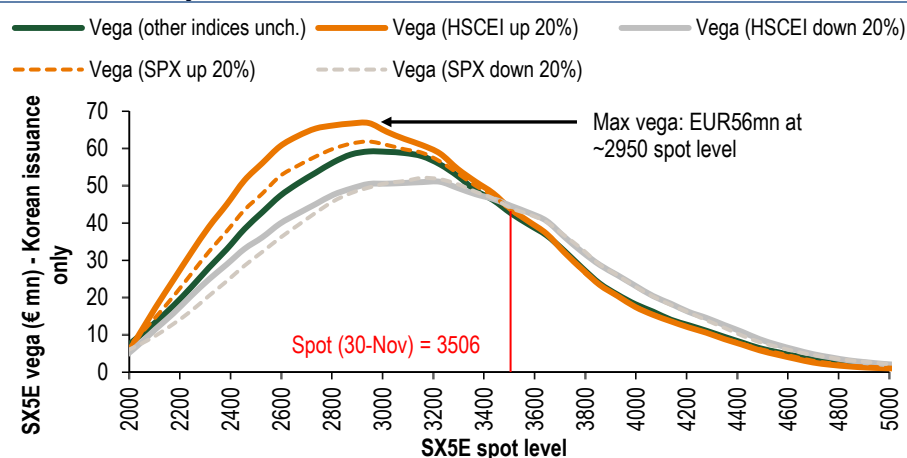
Auto-callable issuance should decline from all-time highs

Korean auto-callables had a record year in 2015 totaling US\$53bn YTD and supplying over US\$200m vega to HSCEI, SX5E, KOSPI2 and SPX. With the lackluster HSCEI performance, many outstanding products (US\$40bn) still have not knocked out early and hence are restraining the new issuance amount. We expect 2016 issuance to fall vs. 2015 but barring any market crash events, issuance size in 2016 should still remain strong as investors are still searching for yield in an easing cycle.

Regulator has put up limit on HSCEI products: Capping new vega supply

Korean regulators have limited new HSCEI linked issuance to no more than 80% of the existing redemption amount. To reduce concentration risk and diversify product range, issuers will continue to look for alternative underlyings such as Nikkei or participation products. For instance, over 20% of the Oct-2015 issuance was not a typical worst-of HSCEI/SX5E/KOSPI2/SPX product, the highest in the last 2 years.

Chart 19: ESTX50 vega outstanding from Korean autocallable issuance for the same levels of the ESTX50 is materially different for +/- 20% moves in the HSCEI



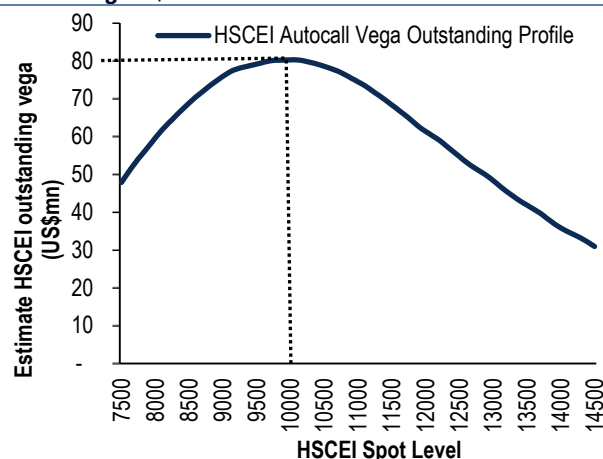
Source: BofA Merrill Lynch Global Research. Data is as of 30-Nov-15

Current peak of issuers' vega: HSCEI 10,000; KOSPI2: 200; NKY: 17,500

Issuers' vega exposure will first peak & then begin to fall as the underlying approaches the knock-in levels. We estimate issuers' vega exposure will peak at 10,000 level for the HSCEI, the 200 level for the KOSPI 200 and 17,500 for the NKY. Current issuers' vega outstanding for HSCEI, KOSPI2 and NKY are US\$80m, US\$29m and US\$26m respectively.

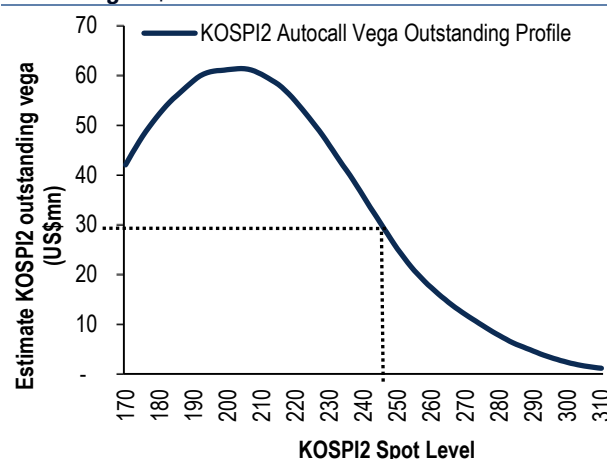
slowed down in 2H2015 due to investors' interest in the Japan Post IPO and the market turmoil. With existing products likely to be knocked-out, we believe issuance should pick up in 1H2016 and put some downward pressure on NKY volatility.

Chart 20: HSCEI vega peak is at 10,000; Current vega outstanding: US\$80m



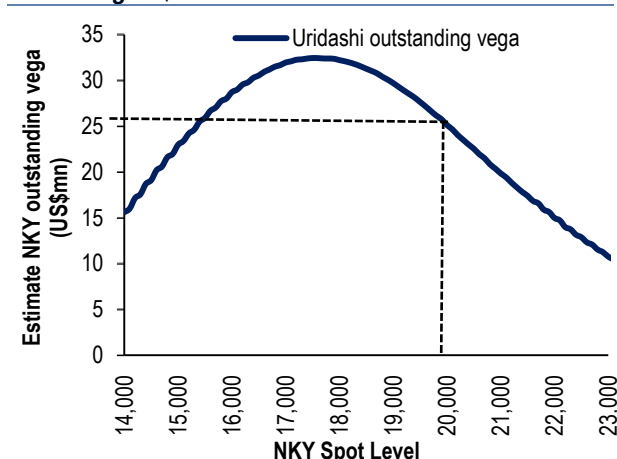
Source: BofA Merrill Lynch Global Research As of 30-Nov-2015

Chart 21: KOSPI2 vega peak is at 200; Current vega outstanding: US\$29m



Source: BofA Merrill Lynch Global Research As of 30-Nov-2015

Chart 22: NKY vega peak is at 17,500; Current vega outstanding: US\$26m



Source: BofA Merrill Lynch Global Research As of 30-Nov-2015

50% of the existing auto-callables will likely knock-out if HSCEI > 10,500

The underlying of a typical worst-of auto-callable is based on the worst performer of a basket of indices. Given its underperformance, the HSCEI has been the key factor in preventing early calls of products. We estimate that 50% of the existing US\$80m HSCEI vega will be knocked-out if the HSCEI stays above 10,500 in 1H16 while 80% of the US\$80m HSCEI vega will be knocked out if HSCEI stays above 12,000.

Negative spot volatility correlation should persist if HSCEI stays < 10,500

Since Aug-15, HSCEI volatility has started to exhibit a negative spot-vol correlation as the HSCEI is trading below the issuers' vega peak (Chart 20). This behavior observed in the HSCEI is also affecting other Asian indices such as KOSPI2 and to a lesser extent NKY. Positive spot volatility correlation may return if the HSCEI stabilizes above the 11,000-12,000 levels combined with new issuance.

Japan issuance has been weak in 2H15 but issuance should pick up in 2016

There has been US\$6.4bn in public NKY Uridashi issuance YTD, which is at a similar level to 2014 but 75% of the products were issued in 1H2015. The issuance trend has

Inter-index correlations elevated due to Korean auto-callable issuance

As long worst-of-puts (the dealers' position) are short correlation, they need to buy inter-index correlation between the popular indices referenced in structured products issued out of Korea, namely ESTX50, HSCEI, SPX & KOSPI (see Chart 17 for an indication of how the popularity of different indices has evolved). While it is difficult to isolate the impact on correlation from structured products alone, as [correlations have structurally risen across asset classes](#), we believe that the structured product impact can be material. Indeed on page 19, we suggest a long best-of call on ESTX50, SPX & HSCEI vs a short HSCEI call, which is attractively priced in part due to elevated inter-index correlation.

Div outlook: favour EU/China pull-to-par

With the launch of SPX div futures and ESTX50 div future liquidity rising by 10% YoY, div derivatives have continued to expand in 2015. Pull-to-par trades offer attractive risk-reward. In particular, HSCEI Dec16 & ESTX50 Dec17 divs offer >10% upside potential vs. our forecast. A pick up in US inflation could support SPX long-term divs.

European dividends: Favour Dec16 & Dec17 for long ESTX50 div exposure

For European dividend upside, we prefer short-dated div futures over long-dated. With access to cheap funding (EU IG spreads ~1%) and low balance sheet leverage of the ESTX50 constituents (total debt/asset ratio = 28%, 14th %ile since Jan-02), the risk of div cuts over the next 1-2 years appears limited in our view. Based on BofAML fundamental bottom-up forecasts of 119.4 & 128.7, we expect 9% & 11% p.a. upside in Dec16 & Dec17 div futures (if unwound in Apr16 & Apr17 respectively, Table 2). Note that Oil & Gas div risk is limited as Total (biggest contributor to ESTX50 divs) [remains committed](#) to scrip divs (to save cash) in '16 & '17 even if oil prices stay low.

No place to hide in long-term ESTX50 divs; structured product impact & low inflation

We estimate low upside in 4y+ div futures (based on our [Top-down div forecasts](#) which are inflation expectation dependent) than in <3yr div futures (vs bottom-up forecasts). In fact, 3-10yr inflation swaps would have to more than double (rise by 150bps) for 4yr+ divs to offer as much upside as Dec17 divs (11% p.a.). Such a scenario is unlikely as per our EU rates strategists, who [doubt](#) ECB QE2 will be able to restore inflation expectations. Even in the unlikely case of 11% p.a. upside, the risk-reward in 4yr+ divs would be inferior (due to higher risk) vs short-dated futures. Moreover, further growth in structured product issuance ([p.12](#)) & dealer hedging will likely weigh on long-dated divs.

US dividends: long-dated divs to be supported if div payouts & inflation rise

The CME launched SPX div futures contracts on 14-Nov-15. SPX div futures are pricing in 2.2% growth p.a which is low vs our 7.4%p.a bottom-up growth forecast until 2020 and vs 9.9%p.a realised div growth since Dec12. Our US strategists [expect](#) 5% EPS growth in

2016 and suggest high balance sheet cash (vs history) can keep divs supported. Div payout ratios may even increase from their historical lows and share buy-backs (which are not accounted for in div futures) may be less favorable in a rising rates environment. Low long-dated div future levels are currently consistent with low inflation expectations. However, our US economist has recently turned more hawkish (on the Fed rate hike debate) as he finally sees a [credible case for higher inflation](#), which should support long-dated SPX div futures in 2016.

HSCEI divs: Value in technically depressed Dec16 divs as an upside hedge for Chinese banks

The HSCEI div futures curve is inverted (downward sloping) given concerns around a Chinese hard landing + bank earnings as well as selling pressure from dealers hedging structured product issuance. As highlighted [recently](#), the 9% YoY decline priced in Dec16 div futures (Chart 25) is as extreme as the 2009 YoY decline in HSCEI divs.

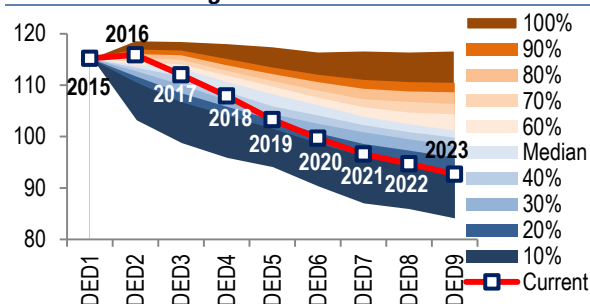
Trade: HSCEI Dec16 div futures could offer attractive risk-reward as an upside hedge for Chinese banks, which are expected to pay 63% of HSCEI dividends in 2016.

Table 2: Most attractive risk reward in ESTX50 and HSCEI short-dated div futures

	ESTX50			HSCEI			SPX		
	Fore-cast**	Div futures	Upside potential*	Forecast	Div futures	Upside potential	Forecast	Div futures	Upside potential
Dec-16	119.5	115.9	8.8%	400.6	366.5	27.5%	46.4	45.0	8.9%
Dec-17	128.8	112.0	10.8%	408.8	366.7	8.3%	49.7	45.9	6.1%
Dec-18	139.1	107.9	11.3%	--	--	--	54.4	46.6	6.8%
Dec-19	137.4	103.3	8.8%	--	--	--	57.8	47.6	6.0%
Dec-20	144.2	99.7	8.8%	--	--	--	61.7	48.1	5.9%
Dec-21	151.5	96.6	8.7%	--	--	--	--	--	--
Dec-22	159.3	94.7	8.5%	--	--	--	--	--	--
Dec-23	167.7	92.7	8.4%	--	--	--	--	--	--

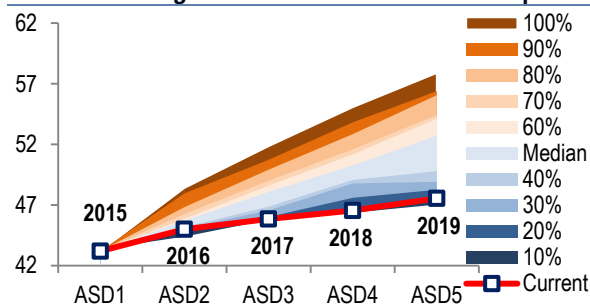
Source: BofA Merrill Lynch Global Research. Data: 4-Dec-15. * Return p.a on a long div trade assuming unwind on 15-April of expiry year ** BofAML bottom-up forecasts, except ESTX50 Dec19-Dec23 forecasts which are top-down (based on inflation swaps, see [here](#)).

Chart 23: ESTX50 long-dated divs are low vs short-dated



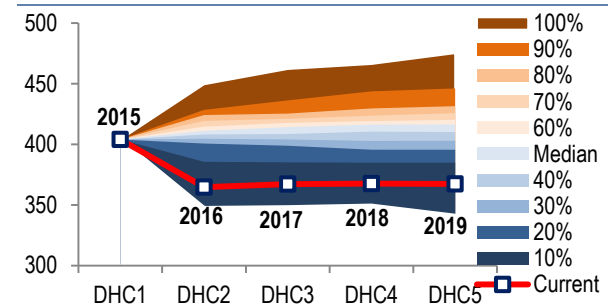
Source: BofA Merrill Lynch Global Research. Data from 4-Dec-12 to 4-Dec-15

Chart 24: SPX long-dated divs low vs historical div swaps



Source: BofA Merrill Lynch Global Research. Data from 4-Dec-12 to 4-Dec-15

Chart 25: HSCEI Dec16 divs are low vs Dec15



Source: BofA Merrill Lynch Global Research. Data from 4-Dec-12 to 4-Dec-15

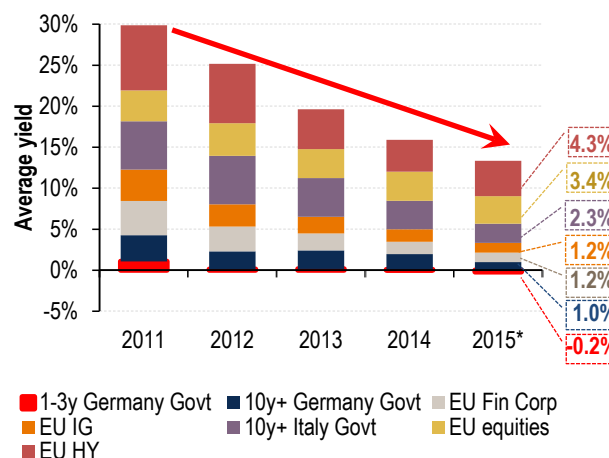
Reaching for yield in Europe given QE

With almost €3trn of negative yielding gov't debt (~45% of the total outstanding), the **hunt for yield** in Europe is as strong as ever. With such low (negative) returns on safe haven assets, central banks have pushed income-seeking investors further out on the risk curve, resulting in a policy-induced contraction in yields across asset classes (Chart 26). With Mario Draghi unleashing QE in 2015 and leaving room to further expand it in 2016, the hunt for yield will likely gather steam, in our view. Indeed, the higher yields on offer in both EU equities and HY credit relative to safe haven assets are among the key investment reasons cited by our [equity](#) & [credit](#) strategists in their 2016 outlooks.

Yield is only half the story: need to weigh vs. asset risk

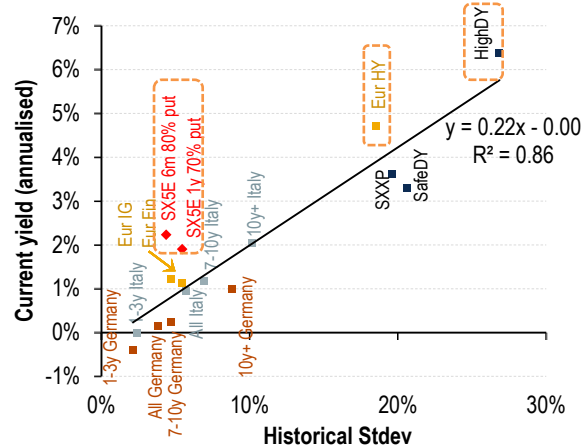
Yield vs risk framework: With a lack of assets offering reasonable yields, investors often focus excessively on (just) the yield on offer with less attention to the associated investment risk. We present a simple yet instructive framework that allows us to assess yield vs investment risk. This compares current yields to historical price volatility (Chart 27) and worst (1y) drawdowns (Chart 28) to measure risk in both the belly & the tail of asset return distributions. The relationship that emerges is intuitively unsurprising – riskier assets offer a higher yield! However, the framework allows us to assess the attractiveness of yield in the context of broad (historical) investment risk.

Chart 26: Central banks have pushed investors further out on the risk curve, which has resulted in a policy-induced contraction in the cross asset yields



Source: BofA Merrill Lynch Global Research. * 2015 corresponds to the YTD average (data ending 04-Dec-15).

Chart 27: Yield vs risk framework: As expected, riskier instruments offer higher yield. Equities and HY bonds screen favorably – consider derivative overlays for additional yield



Source: BofA Merrill Lynch Global Research. Current yield as of 30-Oct-15. Historical stdev is calculated using 1y returns (rolled monthly) from 31-Jan-01-30-Oct15. For 1y puts, 1y return corresponds to the expiry P&L and for 6m puts, we consider 1y return from 2 consecutive held-to-expiry trades. Safe DY & High DY equities are from our quant strategists, as referenced in [Pan European Quant Profiles](#) & [My stock is my bond](#).

European HY bonds & equities stand out: It is interesting to note the dislocation of EU HY bonds in Chart 27 & Chart 28, as they seem to offer superior yields given their historical risk profile (EUR HY above both regression lines). At the same time, given that growth-linked assets like equities present significant capital appreciation opportunity (vs fixed income), it is important to make the case for EU equities too, which screen attractively in this framework particularly given decent upside potential due to inexpensive valuations, [according to our strategists](#).

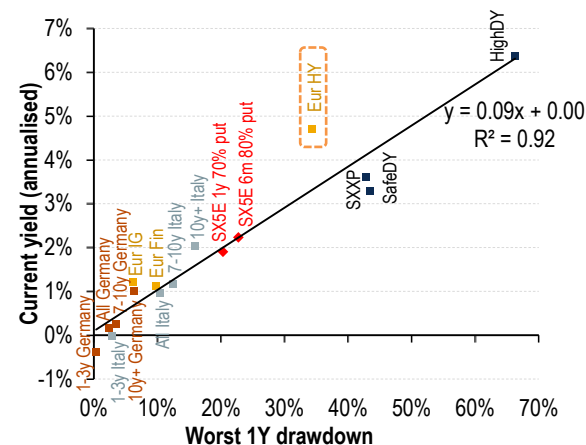
Generate additional yield using equity derivatives

Trade1: Enhance income from liquid optionable names in the EU 'quality' div yield basket by selling 1y OTM covered calls to generate 2.7% additional income on avg

Trade2: Sell ESTX50 1y 70% put (~1.5% premium) – this offers yield at a superior risk-reward vs other fixed income & equity alternatives

Income-oriented investors in pursuit of yield may avoid taking concentrated risk by broadening their opportunity set and creating potential for enhanced returns. To this effect, derivatives overlays can be an important addition to their toolkits. We highlight a few, among several possible ways to generate additional yield via equity derivatives:

Chart 28: Selling OTM ESTX50 puts & call overwriting a div yield basket can help generate additional yield with an attractive risk reward



Source: BofA Merrill Lynch Global Research. Current yield as of 30-Oct-15. Worst 1y drawdown is calculated using 1y returns (rolled monthly) from 31-Jan-01-30-Oct15. For 1y puts, 1y return corresponds to the expiry P&L and for 6m puts, we consider 1y return from 2 consecutive held-to-expiry trades. Safe DY & High DY equities are from our quant strategists, as referenced in [Pan European Quant Profiles](#) & [My stock is my bond](#).

Selling covered calls to boost income from the EU quality div yield basket: Our equity strategists see div yield, specifically sustainable/safe div yield, to remain in demand in 2016. Investors can further enhance their equity portfolio income by selling 1y OTM covered calls. As an example, we consider liquid optionable names from the basket of 'quality' div yield stocks [highlighted by our equity strategists](#). Selling 1y calls on these names with 115% strikes (or 25 delta, whichever is greater) adds ~ 2.7% additional annual income on average to the portfolio (= ~2/3rd of average div yield) while still providing up to at least 15% upside in each stock (see Exhibit 2).

Exhibit 2: Overwriting 'quality' div yield stocks* in Europe (ranked by div yield + call premium)

Ticker	Name	Div Yield	1y Call Strike	1y Call prem	Div yield + call prem
EOAN GY	E.ON	5.5%	127%	4.3%	9.8%
INGA NA	ING	6.5%	116%	3.0%	9.5%
ACA FP	CREDIT AGRICOLE	5.4%	117%	3.2%	8.6%
FP FP	TOTAL	5.5%	115%	3.0%	8.5%
AGN NA	AEGON	4.6%	120%	3.5%	8.1%
BBVA SQ	BBVA	5.1%	116%	2.9%	8.0%
HSBA LN	HSBC	6.5%	115%	1.3%	7.7%
SAND SS	SANDVIK	4.2%	120%	3.5%	7.7%
ERICB SS	ERICSSON	4.4%	118%	3.3%	7.7%
GLE FP	SOC GENERALE	4.4%	119%	3.2%	7.6%

Source: BofA Merrill Lynch Global Research. Data as of 07-Dec-15. Call strike chosen is higher of 1y 25 delta strike and 115%. *Top 10 from the liquid optionable names within the basket of 'quality' div yield stocks [highlighted by our equity strategists](#). For div yield, we use projected 12M div yield using Bloomberg div forecasts.

Selling OTM ESTX50 puts/put spreads: Investors confident of limited equity downside in Europe (due to the ECB put) may consider selling ESTX50 puts/put spreads to generate yield. The risk-reward profile of such structures is exposed to the choice of put strikes & can be tailored to balance upfront income with associated tail risk. In Chart 27& Chart 28, short ESTX50 1y 70% puts appear to offer an attractive yield versus other alternatives even given its higher historical risk.

Selling ESTX50 down-&-in puts: A popular variant to selling puts involves selling down-&-in puts, wherein an investor sells an ATM put that becomes active only when spot goes below the (down) barrier, else the put expires worthless. This allows investors to reduce the likelihood of losses (if a low enough down-&-in strike is chosen) at the expense of bigger losses should they occur. This structure, which is a key component of a typical autocallable product, has been a popular means for (global) retail investors generating yield via European underlyings.

Selling div yield (directly): While the above strategies for enhancing yield rely on monetising volatility risk premium and typically take on short upside/downside tail risk, investors may also sell relatively elevated equity (div) yield by selling div futures & buying equity forwards. Staying with the yield theme, we highlight a twist to the [short ESTX50 div yield trade we suggested earlier](#) by using ESTX50 equity & dividend options (see [pg 27](#) for more).

TOP MACRO/DIRECTIONAL TRADES

Cheap macro directional trades via correlation

1. Long best-of ESTX50, SPX & HSCEI call, short HSCEI call
2. SPX over RUT outperformance call contingent on RUT up
3. Long SPX options contingent on EEM
4. SX5E over SPX outperformance call contingent on SPX up
5. Sell Japan FX/equity correlation for upside

Cheap macro directional trades via correl

Long Best-of ESTX50, SPX & HSCEI call, short HSCEI call

Trade: Long Jun-16 105% best-of call on SX5E, SPX and HSCEI & short HSCEI Jun-16 105% call (equal notional) for **1.9%** (quanto-EUR) or **2.0%** (quanto-USD), indicatively.

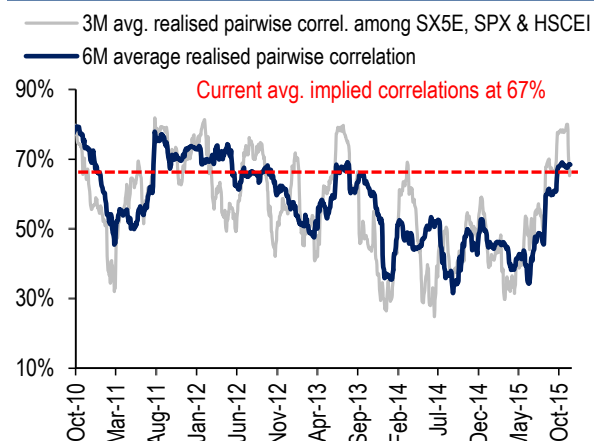
Why? High implied correl. between regional indices provides interesting opportunities for implementing return dispersion if investors share our divergent views on US, EU & China.

Attractive entry point for short inter-index correlation trades: As mentioned on page 14, implied correlations between ESTX50, SPX and HSCEI are elevated due to recent dominance of this index triple in retail structured products. Indicatively, the current average of implied correlation bids is close to historical realised (~67%, which is close to the highest in 5yrs, Chart 29). We believe this is partly due to the willingness of dealers to offload their correlation risk arising from structured products.

Best-of call vs HSCEI call screens well among menu of short correl trades

We consider a menu of seven short correlation trades, which include different combinations of basket calls, worst-of calls, best-of calls & individual calls on the ESTX50, SPX and HSCEI, as well as a call on the return dispersion between these indices.

Chart 29: Avg. realized pairwise correlation between these indices is close to its highest level in 5 yrs.



Source: BofA Merrill Lynch Global Research. Data is from 2-Dec-10 to 4-Dec-15.

Table 3: The long 'Best-of' & short HSCEI call structure screened to be the most attractive with an average payoff/cost ratio of 1.8x and a 90th %-ile payoff/cost ratio of 5.3x

Structure	Price	Correl*	Frequency of profitability	Avg. payoff /cost	90 th %-ile payoff /cost
+Best-of - HSCEI	1.91%	-0.03%	49%	1.8	5.3
+Individual - Worst-of	2.59%	-0.02%	48%	1.2	2.7
+Best-of - Individual	3.84%	-0.03%	41%	0.9	2.0
+Best-of - Basket	4.45%	-0.05%	41%	1.0	2.0
+Individual - Basket	0.86%	-0.02%	35%	1.0	3.1
+SX5E - Worst-of	1.98%	-0.02%	35%	1.4	4.9
+Return dispersion**	2.50%	-0.10%	18%	0.5	1.7

Source: BofA Merrill Lynch Global Research. Data is as of 4-Dec-15. *Correlation sensitivity, i.e. change in %price per correlation point change. **Payoff of 5% call on return dispersion=Max {Avg [($r_1 - r_{avg}$)], -5%, 0}

Based on the historical (5y) performance of these trades at current pricing (Table 3), we prefer long best-of calls, short HSCEI calls, as it lines up favorably under our macro outlook for regional indices in 2016.

EU/US equity markets likely to outperform China: With the US economy staying on track toward a healthy recovery and the ECB continuing to ease, our strategists are [positive on US & EU equities](#). Meanwhile, our economists expect China to [continue on the path of a growth slowdown](#) amid economic structural changes. Also, further potential [CNY devaluation](#) could trigger another Chinese equity sell-off akin to Aug-2015, in our view.

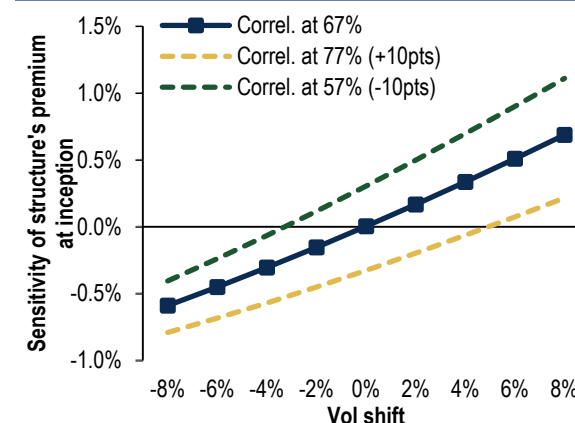
Downside is limited to the initial premium paid: The worst scenario of the trade is a sharp rally in HSCEI while other markets are relatively muted. However, in this case, the long best-of call will have the same payoff as the short HSCEI call, thus limiting the trade's loss to the initial premium paid.

Attractive historic risk-reward: Based on current pricing, the structure would have been profitable ~50% of time in the past 5 years, with an average payout ratio of 1.8x & a 90th %-ile payoff/cost ratio of 5.3x.

Structure entry point benefits from low vol and high correlation

In Chart 30, we highlight the price sensitivity of the long 'Best-of' call / short HSCEI call structure under different correlation and vol scenarios. If vol surfaces shift up by 2 vol pts on all three indices, the structure will be ~20bps more expensive. Meanwhile if implied correlations decrease by 10pts to 57%, the premium will increase by ~30bps.

Chart 30: The long 'Best-of' & short HSCEI call structure benefits from low vol and high correlation



Source: BofA Merrill Lynch Global Research. Data is as of 4-Dec-15. We assume parallel shifts on SPX, SX5E and HSCEI vols.

US large (S&P 500) over small cap (Russell 2k) equities

Trade: SPX over RUT 1Y ATM outperformance call contingent on RUT up at expiry for **0.9%** (~77% discount vs. unconditioned outperformance call; 92% implied correlation)

Why? The trade leverages our US equity strategists' view that we are in the early stages of a [secular trend that will continue to benefit large cap over small cap](#). This is predicated on an expected reversal of the easy monetary policy / high liquidity regime in place since the Tech Bubble, which favoured highly-levered, credit-sensitive, smaller-cap, and lower-quality stocks / sectors relative to their more liquid, larger-cap, and higher-quality counterparts. While some expect small caps to outperform in a rising dollar environment, our strategists note that the historical link is inconsistent at best and is not the primary driver of relative performance between the two size segments.

DM (S&P 500) vs. EM (EEM) equities

Trade 1: SPX 1Y ATM call contingent on EEM < 95% at expiry for 0.86% indicatively (~87% discount vs. vanilla SPX 1Y ATM call; 70% implied correlation)

Trade 2: SPX 1Y 95% put contingent on EEM > 100% at expiry for 0.7% indicatively (~88% discount vs. vanilla SPX 1Y 95% put; 70% implied correlation)

Why? The first trade is consistent with BofAML's preferred [long DM > EM asset allocation](#) for 2016 and suits investors who anticipate a continuation of the past 3 years, when the S&P returned 30%, 11%, and 2% vs EEM returns of -6%, -6%, and -14%. The US economy still appears strongest, and a rising US rates environment is pushing the USD higher and sucking liquidity away from EM. China remains a dark cloud over the EM and commodities complex, with [no signs yet of a turn-around](#).

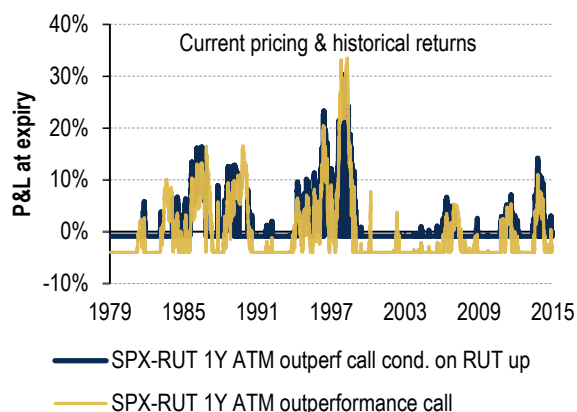
The second is a contrarian trade firmly against the grain of sentiment and positioning in equities/FX. Our Fund Manager Survey shows [EM vs. DM positioning now at the most extreme level on record](#), with EM the biggest short vs. FMS history and China the biggest tail risk on investors' minds. A significant counter-trend rally could easily materialise in EM on a slight change of sentiment. Even if temporary, the resulting dispersion of returns could push the trade firmly in the money.

European (ESTX50) over US (S&P 500) equities

Trade: ESTX50 over SPX Jun16 105% outperformance call contingent on SPX up at expiry for **0.8%** (quanto-€) or **0.9%** (quanto-\$), ~54% discount vs. unconditioned outperf. call; 79% implied correlation)

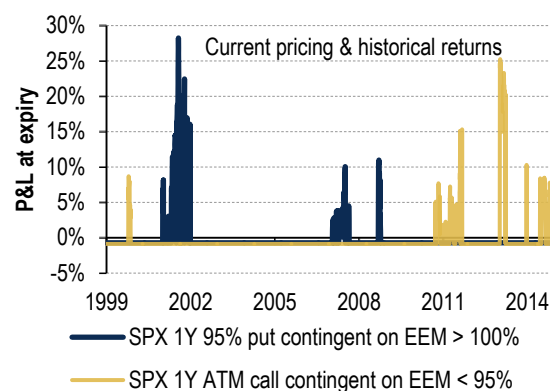
Why? With the ECB unleashing (and expanding) QE in 2015 and the Fed looking likely to begin its hiking cycle on 16-Dec-15, EU equities appear well poised to reverse their multi-year underperformance vs. the S&P.

Chart 31: For ~90bps, investors can position for US large cap vs. small cap outperformance in 2016 (BofAML base case)



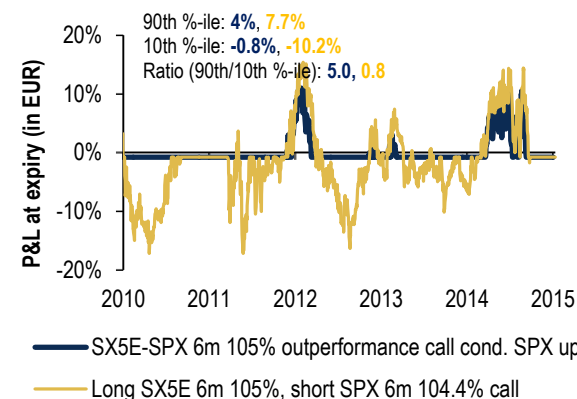
Source: BofA Merrill Lynch Global Research. Data from 27-Dec-79 through 4-Dec-15.

Chart 32: For ~70-90bps, investors can trade either DM > EM (BofAML base case) or EM > DM (contrarian) in 2016



Source: BofA Merrill Lynch Global Research. Data from 20-Dec-99 through 3-Dec-15.

Chart 33: For ~80bps, investors can position for potential out-performance of EU vs. US equities in 2016 (BofAML base case)



Source: BofA Merrill Lynch Global Research. Data from 2-Dec-10 to 4-Dec-15. We strike the SPX call so that the long SX5E/short SPX call premium = that of the outperf. call.

Japan FX/equity correl: sell for upside

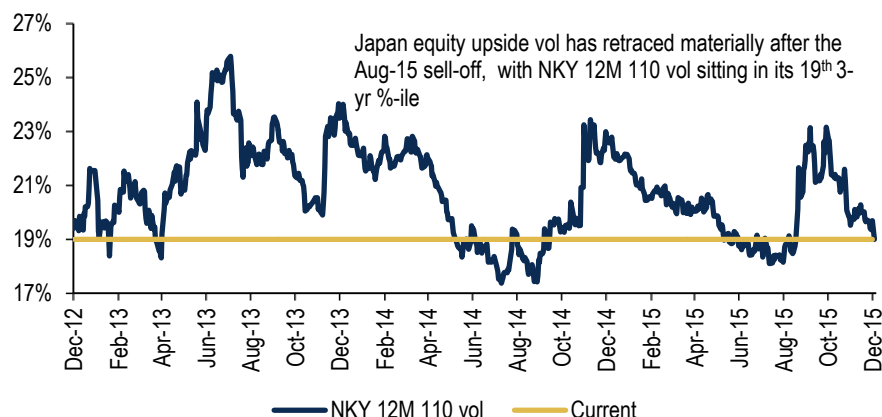
Trade 1: Buy NKY Dec16 105-120% call spread contingent on USDJPY<120 for 1.25% (ref. NKY 19,504, USDJPY 122.8), a 66% discount to a vanilla call spread.

Trade 2: Buy TPNBNK Jun16 105-115% call spread for 2.9% indicatively (ref. TPNBNK 212.5), with 3.4 maximum payout ratio.

Why? A NKY call spread contingent on lower USDJPY provides around a 2/3rds discount to a vanilla call spread – an efficient way of leveraging the diverging long-term prospects of this FX / equity pair. Japanese banks can potentially gain the most from rising asset prices and an improving inflation outlook in the medium term.

Japanese equities still a favourite pick on policy divergence: Going into 2016, central bank policy divergence is undoubtedly among the most important considerations for investors' asset allocation. With inflation still well below the BOJ's target of 2%, there is scope for further monetary and fiscal easing, which in turn supports Japanese equities. Meanwhile, the Fed is poised to [begin a policy normalization process at its December meeting](#). Our global investment strategists [recommend](#) Japanese stocks as one of their 2016 top investment picks. Indeed, our equity strategists also [expect](#) TOPIX EPS to reach record highs in FY3/17 and note that supply / demand factors are currently biased toward the positive, such as the return of foreign investors as buyers.

Chart 34: Japan upside vol has come in significantly after the Aug-15 global selloff, while NKY back-end upside skew is also among the lowest compared to other major indices



Source: BofA Merrill Lynch Global Research Daily data from 4-Dec-12 to 7-Dec-15.

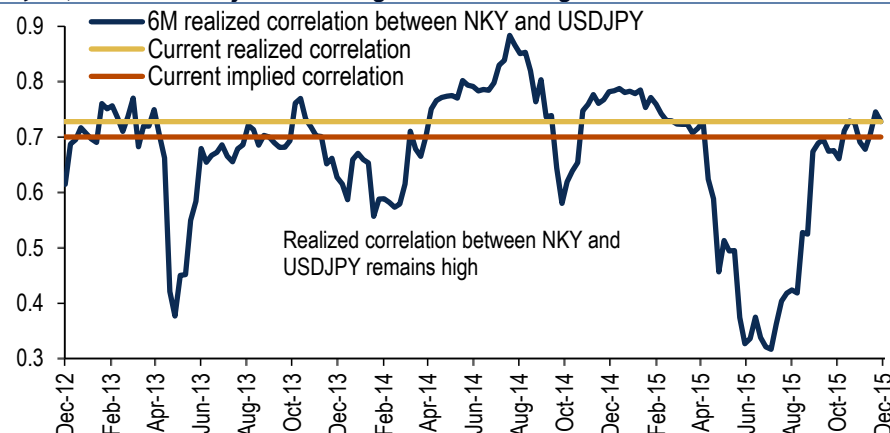
USDJPY faces headwinds after potential additional QQE: Despite potential for further monetary easing, our rates and FX strategists [note](#) that policy sustainability will be questioned after one last monetary and fiscal push in 1Q16, there will be additional headwinds from ECB and PBoC easing, while the yen is already the most undervalued of G10 currencies. Our strategists hence forecast USDJPY at year-end 2016 to be 120.

Buy NKY call spreads contingent on lower USDJPY: Japanese equity implied volatility has materially retraced recently (Chart 34). On the other hand, NKY long-term upside skew is among the lowest in absolute terms of global major equity indices. This makes NKY call spreads attractive for Japanese equity upside exposure.

Investors can further cheapen the call spread structure by conditioning it on lower USDJPY. By entering such a position, investors are effectively short the correlation of the FX / equity pair. Currently, realized correlation between the Nikkei and USDJPY is near YTD highs, pushing implied correlation to as high as 0.70 (Chart 35). The high correlation level marks a particularly attractive entry point for the trade.

Buy TPNBNK call spreads ahead of higher inflation: With potential Fed hiking and further BOJ easing ahead, policy divergence should raise Japan inflation expectations and in turn [benefit](#) the financial sector. As one of the most underweight sectors by Japan equity specialist investors in our [Japan Fund Manager Survey](#) over the past three months, banks have more room to catch up as investors potentially rotate into financials in 1Q16. We recommend investors position for medium-term upside in the sector via TPNBNK call spreads.

Chart 35: Realized correlation between the NKY and USDJPY has risen significantly since tanking in June, and it is currently within striking distance to YTD highs



Source: BofA Merrill Lynch Global Research Weekly data from 29-Jun-12 to 4-Dec-15.

TOP ALPHA/RELATIVE VALUE TRADES

1. Leverage X-market vol dislocations for RV
2. Enhancing 'traditional' S&P500 call write programs by combating upside tail risks
3. Energy vol dispersion too cheap to ignore
4. Sell yield via long EU equity, short divs
5. Cheapen +HSCEI-ESTX50 var relative value

Leverage X-market vol dislocations for RV

Trade: Following a GFSI equity vol dispersion signal (see below), trade the top 3 long / short option pairs (1M 25d puts or calls) as identified by the [X-asset option RV framework](#).

Size for a target income of 1% (alpha profile) by selling 0.66% worth of each short option and spending half the proceeds on the respective long options or, **alternatively size for zero cost** (hedge profile) by trading 0.33% of each long/short option pair.

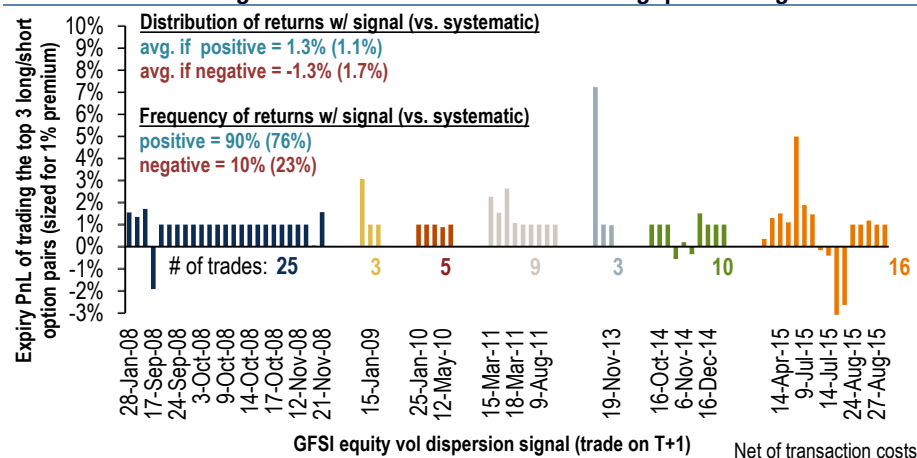
Why? Instances of stark dislocation between X-market vols present attractive entry points for X-asset options RV trades as (i) laggard vols rise in case of broader contagion, or (ii) stressed vols decline as asset-specific risks dissipate. Trading the top 3 option pairs on signal dates would have generated positive returns 90% of the time (alpha profile), while profits would have been >2x as frequent and 4x as large as losses (hedge profile).

Using GFSI to detect cross-market volatility dislocations

As noted in our Outlook, we expect markets in 2016 to be characterized by rising fragility and turbulence. Although trading a less stable world poses many challenges, such shocks can also be a source of opportunity in our view, creating actionable entry points for cross-asset relative value between leader and laggard assets.

We provide proof of concept here based on a signal that triggers when any one of the GFSI equity vol sub-components (five in total spanning SPX, ESTX50, FTSE, HSI, and

Chart 36: The target income case (alpha profile) would have generated positive returns 90% of the time while the average loss would not have exceeded the average profit in magnitude



Source: BofA Merrill Lynch Global Research. Data from 28-Jan-08 (first signal date in 2008) to 28-Aug-15 (latest signal date).

NKY) dislocates sufficiently over a 5-day period vs. the remaining GFSI vol measures (six spanning EURUSD, GBPUSD, USDJPY, USD & EUR interest rates, and broad commodities) and credit spreads (four spanning US & EU IG & HY corporates).

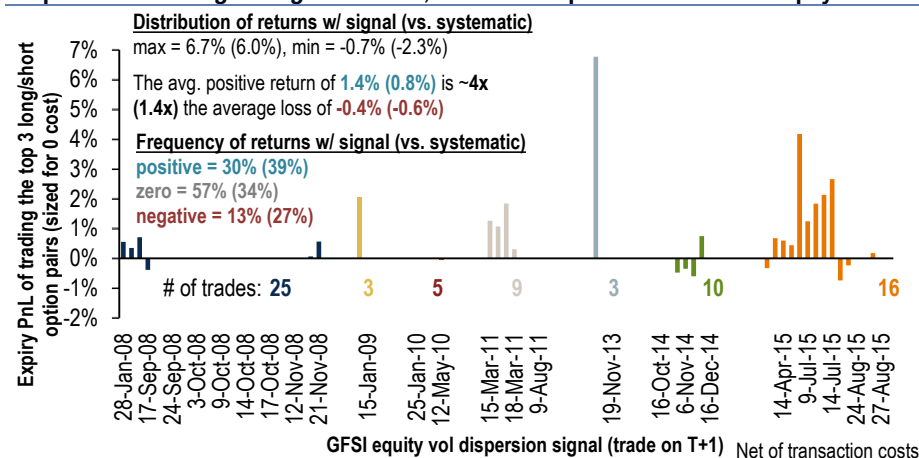
How to leverage extreme volatility dislocations for alpha or zero-cost hedges

In [Alpha from cross-asset option relative value](#) we introduced a framework to identify long/short option pairs (1M 25d puts or calls) which screen attractive from a current pricing vs. historical risk perspective. Systematically selling the top 3 screening short options and using 50% of the premium proceeds to buy the respective long options would have generated 6x the Sharpe of systematically selling S&P puts, since Jan-08.

Using this framework and selecting appropriate long/short ETF option pairs (ETF universe as in the [report](#)) whenever a GFSI equity vol dispersion signal triggers, extreme X-asset volatility dislocations can be monetized tactically. The historical PnL between trade initiation and the subsequent expiry would have been positive 90% of time since '08, with comparably sized profits and losses on average (Chart 36).

Alternatively, sizing the long/short option pairs for zero cost at initiation leverages the high premium received from the short option legs (by buying a larger than usual number of long options), without being net short vol during possibly turbulent markets. This implementation would have resulted in positive returns >2x as frequently and 4x as large as negative returns (Chart 37). We note, however, that for a considerable number of trades both the long and short option positions would have expired worthless (net PnL = 0).

Chart 37: Positive returns in the 0-cost sizing alternative (hedge profile) would have been >2x as frequent and 4x as large as negative returns, but with multiple instances of 0 net expiry PnL



Source: BofA Merrill Lynch Global Research. Data from 28-Jan-08 (first signal date in 2008) to 28-Aug-15 (latest signal date).

Outlook for S&P500 call overwriting: constructive but beware the upside tails

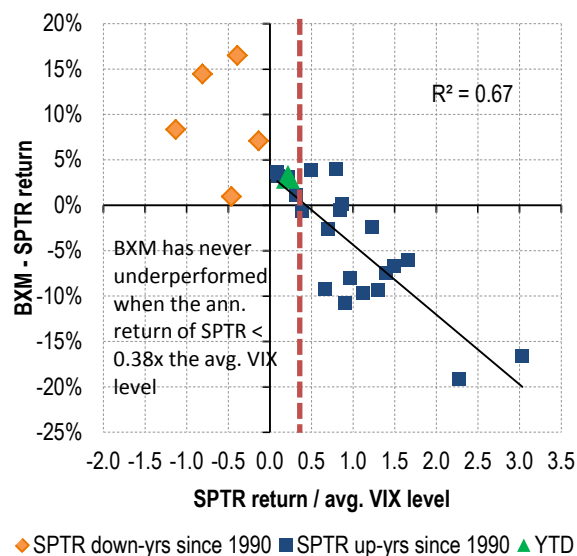
Trade: Enhance S&P500 1M call overwriting programs through weekly options to diversify strike risk and consider reducing risk after sharp market drawdowns.

Why? Our outlook for higher vol coupled with [modest equity returns](#) is supportive of robust performance for covered call strategies in 2016. However, the persistence of [upside tails](#) remains a key threat to the performance of the strategy. We outline below some levers that investors can pull to combat these upside shocks. Selling 4w 25d calls, every week, would have outperformed a 'traditional' front-month 25d call write strategy by 3.6% since Dec-13. Selling 4w 25d-15d call spreads when the 3M equity drawdown is larger than the short 25d call breakeven would have further improved the performance of the strategy by 2.9%.

Constructive outlook predicated on modest S&P returns and higher implied volatility: As seen from Chart 38, the S&P500 return relative to implied volatility well explains the return of a systematic 1M call overlay program (e.g., BXM) relative to the equity market. Since 1990, so long as the ann. total return of the S&P has not exceeded ~0.38x the average VIX level for the year, BXM has never underperformed. In other words, overwriters expecting [~7% total return for the S&P in 2016](#) (5% price return + 2% dividend yield) should wish for an average VIX of at least ~18.5. Of course, these should be viewed as rough estimates, given the path dependency embedded in call overwriting.

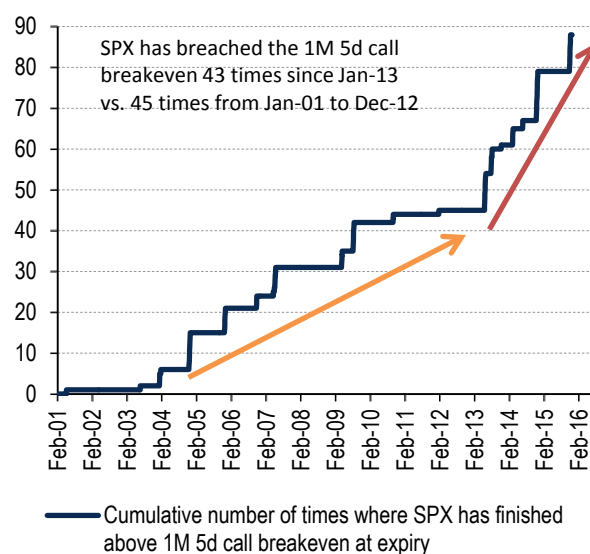
Mispricing of S&P500 upside tails, a key threat to call write performance: Over the past 3 years, the S&P option market has persistently 'underpriced' the ability of equities to rebound violently post aggressive sell-offs. In fact, Chart 39 shows that the breakeven of an S&P 1M 5d call would have been breached nearly as many times since Jan-13 as in the previous 12 years. Since 2001 (and excluding '09 when the market bounced off 'US credit crisis lows') the mispricing of S&P upside tails has never been as extreme as during the past three years (Chart 40). Hence, we view the persistence of upside shocks as a key potential headwind to the performance of call overwriting in 2016.

Chart 38: Since 1990, so long as the annualized total return of the S&P500 has not exceeded ~0.38x the average level of VIX, BXM (1M ATM overwrite) has never underperformed SPTR



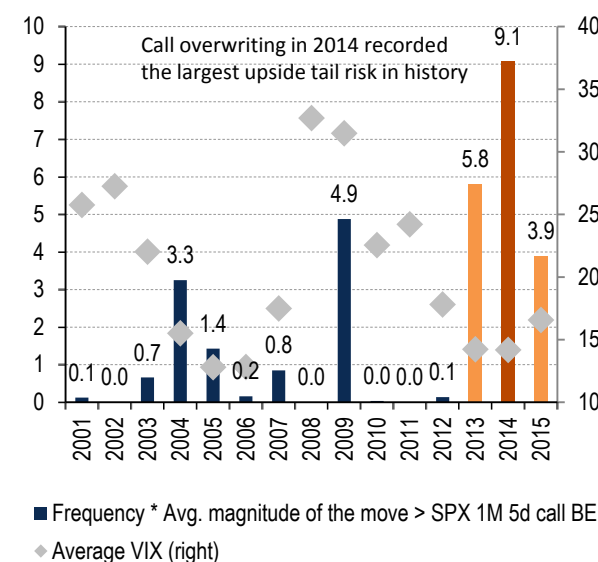
Source: BofA Merrill Lynch Global Research. Data as of 4-Dec-15.

Chart 39: The S&P500 would have breached the upside breakeven of a 1M 5d call nearly as many times since Jan-13 as in the previous 12 years



Source: BofA Merrill Lynch Global Research. Using estimated daily prices and strikes of SPX 1M 5d calls and subsequent SPX 21d returns from 2-Jan-01 to 19-Nov-15.

Chart 40: The 'mispricing' of SPX upside tails by the option market has been extreme over the past three years and could continue to be a threat for covered call programs



Source: BofA Merrill Lynch Global Research. We compute for each calendar year since '01, using daily data, the product of (frequency with which the SPX has breached the breakeven of a 1M 5d call) x (average magnitude of the move beyond the breakeven).

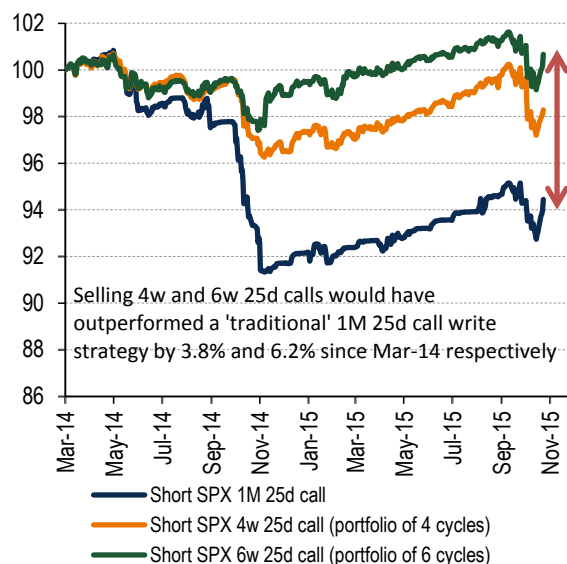
Enhancing 'traditional' S&P500 call write programs by combating upside tail risks

In what follows, we outline some levers that investors can pull to combat upside shocks and enhance the performance of 'traditional' 1M S&P covered call programs.

Diversify path dependency risk with weekly options: As shown [previously](#), incorporating weekly options can materially enhance the risk-return profile of traditional front-month option selling strategies. This is primarily because re-striking a portion of the notional on a weekly rather than a monthly basis can materially reduce the performance sensitivity of the call write strategy to a specific entry point and mitigate the drift in delta of the option overlay as spot moves away from the strikes. As seen in Chart 41, selling S&P500 4w 25d calls every week on 1/4th of the notional would have materially outperformed selling front-month 25d calls on the full notional since Mar-14.

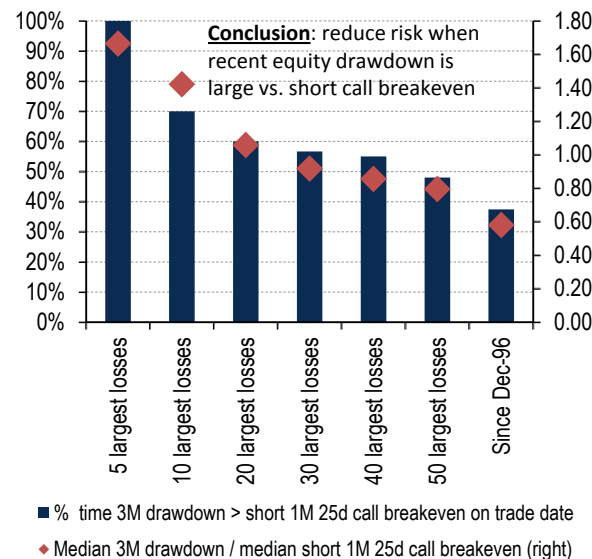
Tenors greater than 1M can offer value depending on market path: Increasing the tenor of the strategy is another sensible approach to reduce the risk of suffering 'intrinsic hits' during upside tails. This comes at the expense of less income generation if the underlying expires below the short call strike (unless more leverage is employed).

Chart 41: Levering the flexibility of weekly options to reduce the impact of upside shocks would have enhanced the performance of traditional short SPX 1M 25d call strategies



Source: BofA Merrill Lynch Global Research. Daily data from 21-Mar-14 to 12-Nov-15. Note that the CBOE started quoting six consecutive SPX weekly options in early 2014.

Chart 42: The worst losses of an SPX 1M 25d call write strategy occurred after equity drawdowns, when the size of the sell-off was larger than the call breakeven post drawdown



Source: BofA Merrill Lynch Global Research. Monthly data from Dec-96 to Oct-15. The 3M drawdown is the current to rolling 3M peak level of S&P500 return on a trade date.

For instance, Chart 41 shows that selling 6w 25d calls, on a weekly basis, would have outperformed selling 4w 25d calls since Mar-14. Hence, slightly longer-dated structures than 1M were more immune to the occurrence of upside shocks and better suited to the market's pace of mean reversion in recent periods.

Reduce risk exposure when equity drawdown is large relative to short call breakeven: Given the severity of the 'mispricing' of upside tails in the past 3 years, overwriters should also consider reducing their risk exposure when implied volatility appears too cheap to protect the strategy against equity market snap backs. As seen from Chart 42, the worst losses of a systematic 1M call write strategy typically occurred in months where the magnitude of the recent equity drawdown was materially larger than the short call breakeven. Chart 43 shows that not selling calls or selling 4w 25d-15d call spreads when the 3M market drawdown is larger than the short 25d call breakeven would have materially outperformed systematically selling 4w 25d calls over the past two years. This outperformance of the 'dynamic' call write strategy in mean-reverting markets comes at the expense of less income generation in periods where the equity market repeatedly finishes below the call strike (e.g., protracted sell-offs).

Chart 43: Selling call spreads / going into cash when the recent 3M drawdown > call breakeven would have enhanced the performance of an SPX 4w 25d call write strategy



Source: BofA Merrill Lynch Global Research. Daily data from 31-Dec-13 to 12-Nov-15. We assume transaction costs of 2.5bps of the SPX level to initiate each call.

Energy vol dispersion too cheap to ignore

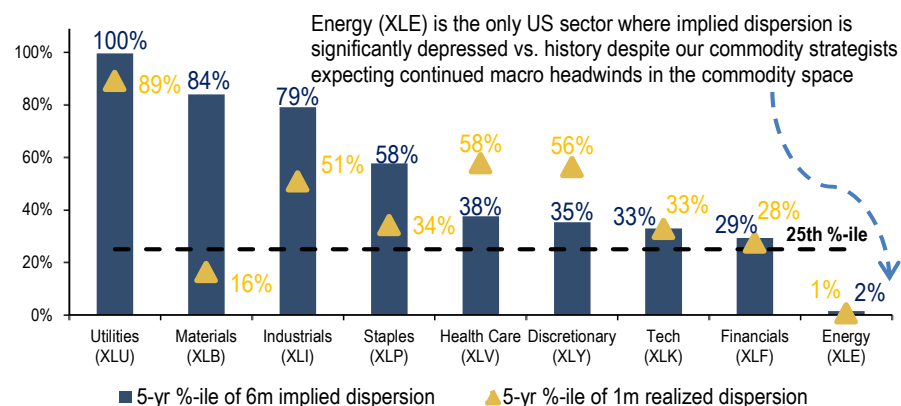
Trade: Long Top11 XLE 6m volatility dispersion gamma sized for indicatively 16.2 vol⁵
Why? Energy intra-sector dispersion is extremely depressed vs. other sectors/its own history. Owning Energy dispersion is likely to profit from further potential commodity weakness while carrying positively in nearly all scenarios realized since Jan-14.

Disconnect vs. other sectors is not reflective of fundamental risks: Energy dispersion is trading at strikingly depressed levels vs. other sectors (Chart 44) and vs. its own history (i.e., 5-yr lows). The decline was caused by a drop in short dated realized dispersion as single stock realized vol did not react to recent market choppiness.

Despite this, our commodity strategists [see](#) two major downside risks within the commodity space: i) CNY currency dropping below our [FX team's projections](#) (CNY at 6.9) and/or Saudi Arabia de-pegging the Saudi Riyal causing oil to collapse to \$25/bbl (a "black swan" scenario). We view these risks as conducive to vol dispersion strategies that are long vol on Energy names and short (a smaller amount of) vol on XLE.

Scenario analysis shows that negative outcomes are highly unlikely: Table 4 illustrates a scenario analysis that can help assess the risk reward of a dispersion strategy that

Chart 44: Energy stands out vs. other sectors with dispersion both on an implied and realized basis standing at its lowest levels in the last five years despite continued macro headwinds ahead



Source: BofA Merrill Lynch Global Research. Daily data from 2-Dec-10 to 2-Dec-15. Single stock baskets represent the smallest baskets needed to capture at least 65% of market cap for each respective sector.

⁵ The actual implied level at which XLE dispersion can be initiated will differ from this, owing to the instruments used (capped variance swaps), tenor (Jun-16 vs. 6m), and bid-ask.

buys vol on a sample equal weight basket of the 11 largest Energy stocks (selected to capture at least 65% of the sector's market cap). Each entry in the table shows the expiry P&L for a particular correl/XLE vol range combo and (in parenthesis) the respective historical probability for that range combo since Jan-14.

- Positive carry expected in status quo:** Were 6m XLE realized vol and realized correl to remain near current levels of 27.3 vol pts and 69.7 correl pts (highest levels since Jan-14, and above their 2H14 'Oil rout' highs), P&L would be positive. Importantly, if correl declined to levels in between 0.55 and 0.60 (which has occurred 71% of the time conditional on XLE staying elevated), the trade would generate an even larger positive P&L
- If 6m XLE realized vol declines to somewhere around 25.0 vol pts (80th %-ile since Jan-14) correlation would have to reach ~0.71 for the P&L to start turning negative, a level last seen in Mar-09
- If XLE vol was to retrace to levels around its median levels since Jan-14 (i.e., 17.6 vol pts) it would take realized correlation to be at least 0.55 for the P&L to be negative. However, conditional on XLE vol being around its median, correlation has never been higher than 0.51 over the past 10 years

Table 4: Scenario analysis for a Top11 XLE dispersion strategy (in \$ thousands)

		6m Realized Correlation (correl pts)									Exp.
		$\rho \leq 25$	$25 < \rho \leq 35$	$35 < \rho \leq 40$	$40 < \rho \leq 45$	$45 < \rho \leq 50$	$50 < \rho \leq 55$	$55 < \rho \leq 60$	$60 < \rho \leq 65$	$65 < \rho \leq 70$	P&L
6m XLE Realized Volatility (vol pts)	$26 \leq \sigma$	\$588 (0)	\$438 (0)	\$288 (0)	\$217 (0)	\$161 (0)	\$116 (0)	\$79 (71)	\$48 (0)	\$21 (29)	\$62
	$24 \leq \sigma < 26$	\$537 (0)	\$398 (0)	\$259 (0)	\$194 (0)	\$143 (0)	\$101 (0)	\$66 (67)	\$38 (2)	\$13 (30)	\$49
	$22 \leq \sigma < 24$	\$441 (0)	\$323 (0)	\$206 (0)	\$151 (0)	\$107 (0)	\$72 (4)	\$43 (47)	\$18 (49)	-\$2 (0)	\$32
	$20 \leq \sigma < 22$	\$353 (0)	\$255 (0)	\$157 (0)	\$111 (0)	\$75 (22)	\$46 (57)	\$21 (22)	\$1 (0)	-\$17 (0)	\$47
	$18 \leq \sigma < 20$	\$273 (0)	\$193 (0)	\$113 (0)	\$75 (0)	\$46 (41)	\$21 (59)	\$2 (0)	-\$15 (0)	-\$29 (0)	\$31
	$16 \leq \sigma < 18$	\$201 (0)	\$137 (0)	\$73 (0)	\$43 (19)	\$19 (79)	\$0 (2)	-\$16 (0)	-\$30 (0)	-\$41 (0)	\$23
	$14 \leq \sigma < 16$	\$137 (0)	\$87 (0)	\$38 (9)	\$14 (91)	-\$4 (0)	-\$19 (0)	-\$32 (0)	-\$42 (0)	-\$51 (0)	\$16
	$12 \leq \sigma < 14$	\$81 (5)	\$44 (91)	\$7 (4)	-\$11 (1)	-\$25 (0)	-\$36 (0)	-\$46 (0)	-\$53 (0)	-\$60 (0)	\$44

Source: BofA Merrill Lynch Global Research. Data from 2-Jan-14 to 2-Dec-15. 20K vega is bought on single stocks. The trade is sized so that index vega sold = (index vol / average single stock vol) * single stock vega (i.e., gamma flat). Numbers in parenthesis represent the historical prob. (rounded and in %-age) since Jan-14 of a particular correl/index vol range combo, e.g., the prob. of $0.65 < \text{correl} \leq 0.70$ conditional on $12 \leq \text{XLE vol} < 14$ vol pts has been 0 (bottom rightmost entry). *Expected P&L= historical prob. * P&L for one XLE vol range across all correl ranges considered, e.g. expected P&L, conditional on $12 \leq \text{XLE vol} < 14$ vol pts, is \$44K (bottom entry).

Sell yield via long EU equity, short divs

Trade 1: Buy ESTX50 Dec18 bullish risk reversal: -3100 (ATMf) put/+3400 (~107%fc) call (90d) for 4.5% credit, sell Dec19 div future (DEDZ9) (equal notional, refs: 4-Dec)

Trade 2: Buy ESTX50 Dec18 -3100/+3400 bullish risk reversal / sell ESTX50 Dec19 125 dividend call (25d) & generate 7% upfront credit (equal notional, refs: 4-Dec)

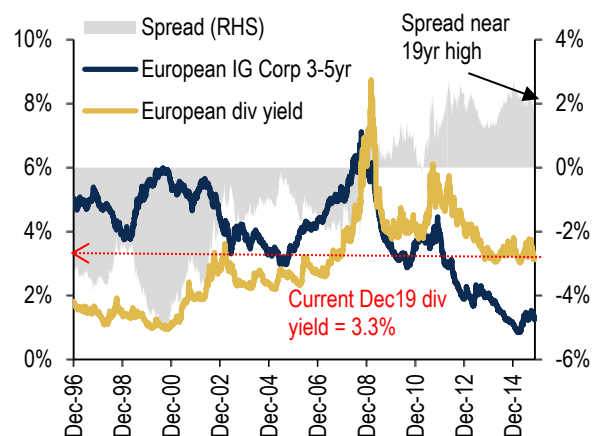
Why? Investors are likely to turn to EU equities as ESTX50 div yield is at 19yr high vs IG bond yields. Sell long-dated divs vs long equity to benefit from (i) down-beta > up-beta & (ii) selling pressure on long-dated divs due to structured products.

Sell dividend yield via long equity forwards/short div futures

As discussed in the Outlook section (p. 16), div yields are likely to compress as investors are likely to turn to EU equities in their search for yield (ESTX50 div yields are at a 19yr high versus European IG bond yields, Chart 45). We recently [suggested](#) buying forwards and selling long-dated div futures on the ESTX50 (and potentially [rebalancing](#) the position). Indeed 3yr+ div futures are likely to face additional downside pressure from dealers hedging their div exposure from ESTX50-linked structured products. We now provide option alternatives for potentially higher yield & partial downside protection.

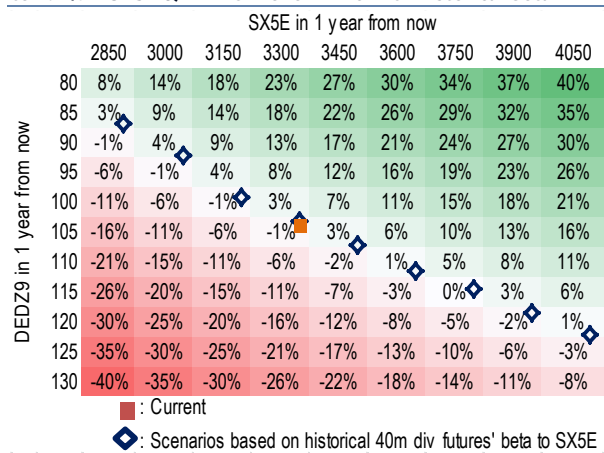
For additional downside protection, consider ESTX50 risk reversals (instead of forwards): We suggest selling an ESTX50 Dec18 3100 (ATMf) put and buying a 3400 call to keep downside protection (at maturity) of ~12% below spot (7.3% put strike + 4.5% credit).

Chart 45: ESTX50 div yields are very high vs Credit IG bond yields



Source: BofA Merrill Lynch Global Research. Data from 31-Dec-96 to 04-Dec-15. Div yield = trailing 12m dividends / equity index. GDP weighted average of Spanish, Italian, German, French div yields pre Jan02.

Chart 46: We estimate the P&L of trade 1 will range from 0.4% to 2.4% if SX5E & DEDZ9 move in line with historical beta



Source: BofA Merrill Lynch Global Research. Data as of 4-Dec-15. Assuming bid-offer of 30bps on DEDZ9 and SX5E Dec18 risk-reversal. Betas used to calculate DEDZ9 performance: 1.0 (if SX5E < -10%); 0.8 (if -10% < SX5E < +10%) and 0.7 (if SX5E > +10%).

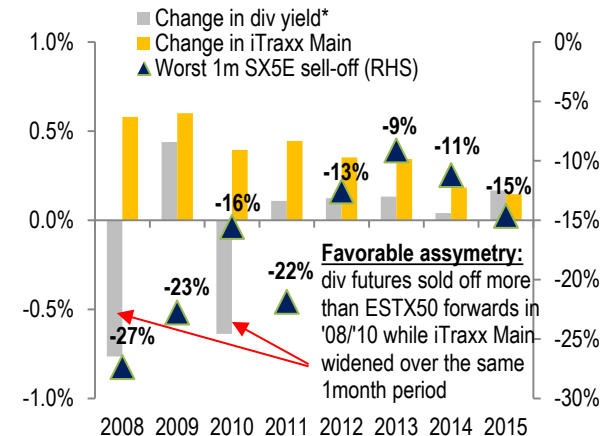
High down-, low up- beta of long-dated div futures provides favourable asymmetry: In [Discovering derivatives can pay dividends](#), we have highlighted that ESTX50 long-dated div futures have an ESTX50 down beta greater than 1 while they underperform the ESTX50 on the upside. As a result, we estimate the P&L of trade 1 is likely to be positive 1 year from now if DEDZ9 & SX5E move in line with their historical beta. If this happens, we estimate the P&L would range from 40bps to 2.4% if the ESTX50 is between 2850 (-17%) and 4050 (+17%) 1 year from now (Chart 46).

For potentially higher yield, sell DEDZ9 125 call against ESTX50 risk reversal: The current DEDZ9 future / ESTX50 Dec18 forward ratio stands at 3.3%. To reduce div risk by allowing room for a long-dated div rally, we recommend investors sell a DEDZ9 125 call vs a bullish ESTX50 risk reversal (trade 2) to generate 7% upfront premium. In case of an equity rally, it is unlikely the trade will experience losses as the div and equity call strikes imply a yield of ~3.7%, which is high vs history (81st %ile since Jan-97 whenever 1yr ESTX50 returns > 0).

Consider iTraxx Main bearish risk-reversals for protection against div yields rising

BofAML credit strategists [anticipate](#) European IG spreads to widen given (i) mediocre inflows in 2H2015, (ii) Reverse Yankees may create supply shocks & weigh on investor sentiment and (iii) liquidity is poor. They [recommend](#) buying payers, selling receivers on iTraxx Main to buy credit protection. Chart 47 illustrates that the CDS index would suit well to hedge long SX5E/short div trades as the CDS index has widened more than the rise in div yields during the worst ESTX50 1m sell-offs since 2008.

Chart 47: iTraxx Main CDS spreads have widened more than the rise in ESTX50 div yield in the worst 1m sell-offs since 2008



Source: BofA Merrill Lynch Global Research. *Div yield = 40m ESTX50 div future / 28m ESTX50 equity forward.

Cheapen +HSCEI-ESTX50 var rel value

Trade: Sell ESTX50 Dec16 VarSwap at 24.8% (indic.) & buy HSCEI Dec16 var replication (8 options as in Chart 48 + a dynamic position in HSCEI) for an implied strike of 27.3%.

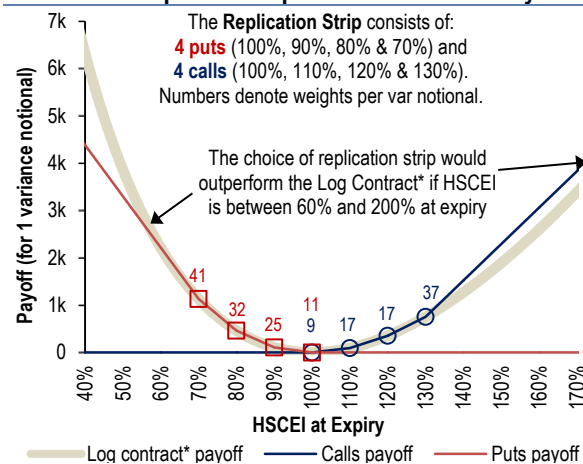
Why? Fundamentally, we like buying HSCEI var financed by short ESTX50 var given our macro outlook (China risk & ECB support). However, as the VarSwap strike spread is structurally rich we prefer buying a HSCEI var replication portfolio vs. the ESTX50 VarSwap, instead. This cheapens the strike spread from 7.4% to 2.5%, making the trade profitable 73% of the time since Nov-05, at current pricing.

Aided by supportive ECB, we expect ESTX50 realised vol to be relatively contained in 2016, absent any spillover from global markets. Meanwhile, as the Chinese govt. attempts the difficult task of managing a slowing economy, spikes in HSCEI realised vol cannot be ruled out. However, extreme HSCEI sell-offs (>40%) are unlikely as China has plenty of savings & policy tools to [stimulate the economy & contain any crisis](#).

Both VarSwap strikes are structurally high; Pair short ESTX50 var with HSCEI var replication:

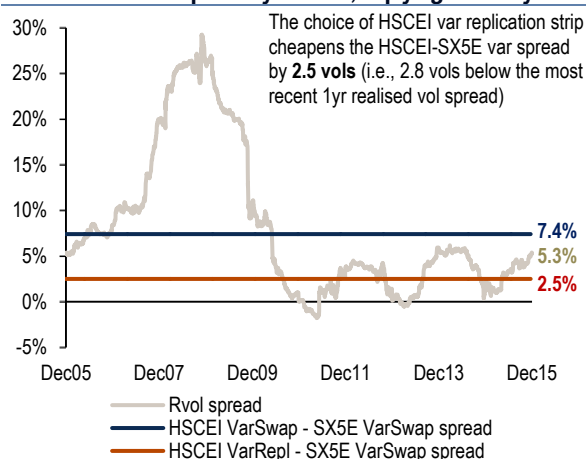
As noted in [More to variance swaps than meets the eye](#), demand for long ESTX50 variance has put upward pressure on the tail liquidity risk premium (TRP) embedded in quoted VarSwap levels, benefiting the entry point of short ESTX50 long-dated VarSwaps, all else equal. The HSCEI TRP is exacerbated by a trio of illiquid options, institutional var demand & retail option supply via structured products.

Chart 48: The replication portfolio payoff would be greater than the VarSwap if HSCEI expires within 60%-200% by Dec16



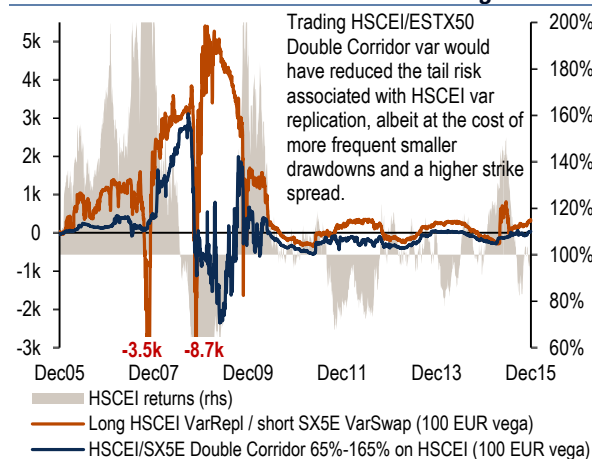
Source: BofA Merrill Lynch Global Research. *Related to the VarSwap payoff (see [here](#)).

Chart 49: Replicating the long HSCEI var leg improves the ESTX50-HSCEI var spread by 4.9 vols, implying +ve carry



Source: BofA Merrill Lynch Global Research. Data from 4-Dec-05 to 4-Dec-15.

Chart 50: The HSCEI var replication trade is short tail risk; trade the 65%-165% Double Corridor on HSCEI to mitigate



Source: BofA Merrill Lynch Global Research. Data from 4-Dec-05 to 4-Dec-15.

TOP HEDGES

1. S&P500 enhanced all-weather hedge
2. Buy best-of-put on crowded FANG stocks
3. Low cost, long vol convexity hedge in VIX
4. China upside and downside hedges
5. KOSPI forward vol is cheapest globally

S&P500 enhanced all-weather hedge

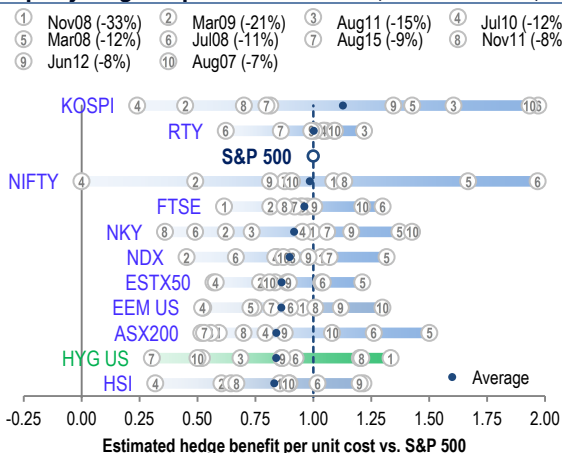
Enhanced All-Weather S&P500 hedge: Buy an S&P 3M 40delta / 25delta put spread & use the proceeds from the short S&P 25d put to buy 3M 25d puts on the top 2 S&P proxy hedges, based on our [Cross Asset Hedging](#) framework (in equal USD sizing)

Why? By leveraging dislocations in cross-asset volatility to buy cheap proxy hedges and selling abnormally rich S&P skew, the All-Weather hedge outperformed the S&P 40 delta put 93% of time since 2008, and generated excess alpha and lower drawdowns versus not only the standard S&P hedged portfolio, but even the unhedged portfolio during 2015.

As equity markets attempt to navigate a landscape of increasingly divergent CB policy in DM, reduced liquidity and lackluster global economic growth, local shocks are likely to persist in 2016, while more pronounced sell-offs (like the one witnessed in Aug-15) also cannot be ruled out. While tactical hedging may well be prudent, the inevitably higher implied vol around such episodes makes outright hedges expensive.

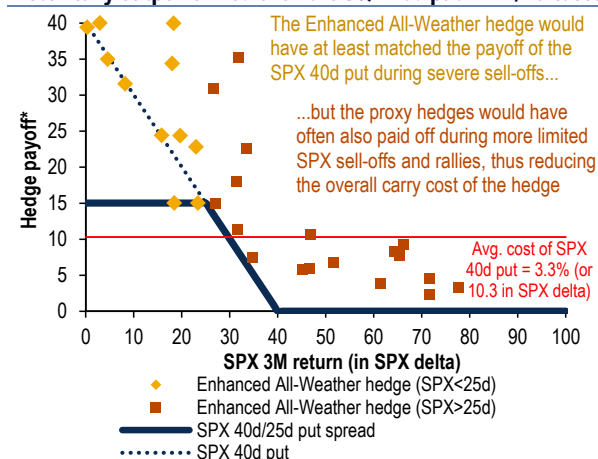
Benefit from historically rich S&P skew to fund cheap proxy hedges: The Enhanced All-Weather hedge combines a 'core hedge' (S&P 3M 40d/25d put spread) with a relatively cheap 'proxy hedge' (top 2 ranking S&P proxy hedges – currently KOSPI & RTY puts, Chart 51) benefitting from historically steep S&P skew (90th 7yr %ile).

Chart 51: KOSPI & RTY puts screen as the most attractive S&P 3M proxy hedges as per our framework* (as of 27-Nov-15)



Source: BofA Merrill Lynch Global Research. Data as of 27-Nov-15. *For details on how to read the chart see our latest [Cross Asset Hedging](#) report.

Chart 52: The Enhanced All-Weather hedge payoff would at least match that of an S&P 3M 40d/25d PS by design, but has historically outperformed even the S&P 40d put in 27/29 cases



Source: BofA Merrill Lynch Global Research. Data: 17-Jan-08 to 27-Nov-15. *For details on how we represent payoffs and premium in delta terms, see [here](#).

By construction, the suggested hedge would at least provide the payoff of an S&P 40d/25d put spread, limiting the possibly adverse effects of the basis risk associated with proxy hedging.

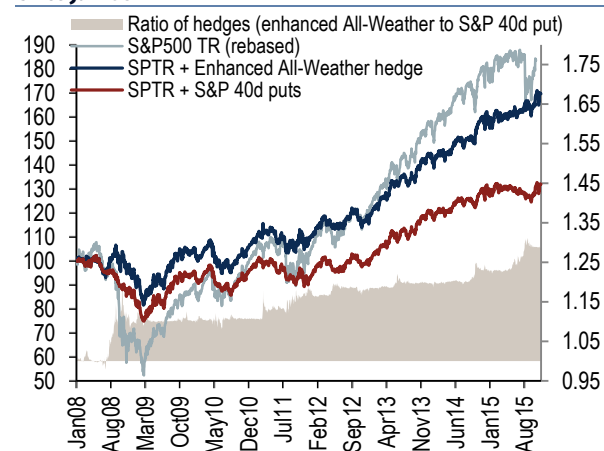
High X-asset correlation during severe risk-off episodes means proxy hedges tend to pay off when needed the most: The Enhanced All-Weather hedge would have at least matched the pay-off of S&P 40d puts during 9 out of 11 large S&P sell-offs (<25d strike level), as risk assets tend to sell-off broadly in tandem during stressed periods (Chart 52).

Improved carry costs in benign markets: Assets occasionally experience idiosyncratic shocks which fail to spill-over into the broader market. In fact, proxy puts paid out 18 times when the S&P was flat or up, reducing the average hedge carry cost (Chart 52).

The Enhanced All-Weather hedge would have outperformed S&P puts & the unhedged S&P TR over multiple time horizons, generating superior risk adjusted returns (Info ratio since Jan-08 = 0.54 vs 0.47 for S&P TR and 0.39 for S&P TR + 40d puts) and reducing the max drawdowns incurred (24% vs 52% for S&P TR and 27% for S&P TR + 40d puts). Improved hedge performance also ensued over the last 1-3yrs, as seen in Chart 53.

And would have even generated alpha in 2015: The Enhanced All-Weather hedge overlaid on the S&P TR would have generated greater returns (6.6%) than the unhedged S&P500 TR index (4.3%), with considerably lower standard deviation (10% vs 15%).

Chart 53: The Enhanced All-Weather hedge outperformed S&P 3M puts & the unhedged S&P TR over multiple time horizons since Jan-08



Source: BofA Merrill Lynch Global Research. Daily data from 17-Jan-08 to 27-Nov-15. *1/3 of the notional is traded once a month. Transaction costs not included.

Buy best-of-put on crowded FANG stocks

Trade 1: Long 1yr 90% Best-of-put on FB, AMZN, NFLX, GOOGL ('FANG') for ~1.85% (~50% implied correlation, ~81% discount to average 1yr 90% vanilla stock put).

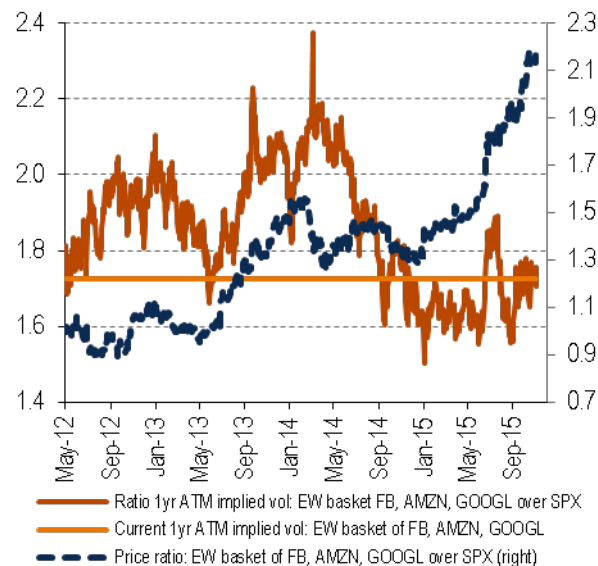
Trade 2: Long 1yr 90% Best-of-put on FB, AMZN, GOOGL for ~2.28% (~52% implied correlation, ~72% discount to average 1yr 90% vanilla stock put).

Why? A 1yr 90% Best-of-put on FANG currently costs ~54% less than a 1yr 90% S&P put, while the last time we saw signs of [rotation out of secular growth stocks](#) in Mar-Apr'14, the best performing FANG stock dropped 14.3%, 3.6x the decline in S&P. The risk of a larger coordinated sell-off is higher today given recent outperformance.

Fears of overvaluation + crowded positioning = higher risk of coordinated sell-off in FANG

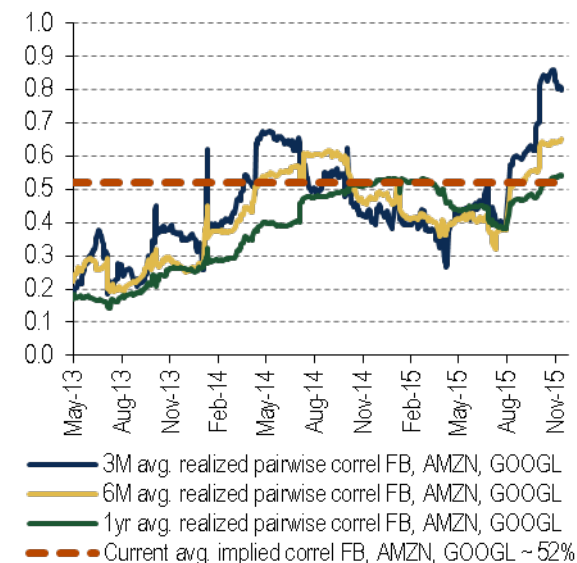
One of the most popular stories in 2015 has been the dazzling performance of fast-growing tech companies such as Facebook (~+36%), Amazon (~+117%), Netflix (~+168%) and Google (~+47%). These four stocks ('FANG') have added \$635bn to the S&P500 market cap vs. -\$434bn for the rest of the stocks, single-handedly carrying the

Chart 54: The average implied vol of FB, AMZN, GOOGL is historically low vs. S&P despite their stellar outperformance



Source: BofA Merrill Lynch Global Research. Daily data from 29-May-12 to 3-Dec-15.

Chart 55: The 1yr implied correlation of FB, AMZN, GOOGL is trading at a material discount to short-term realized



Source: BofA Merrill Lynch Global Research. Daily data from 29-May-12 to 3-Dec-15.

index to its small YTD gains (~+1.59%). This surge in valuation combined with signs of [overcrowding](#) has prompted fears of a correction amongst US equity investors who have often been 'forced' into these stocks to generate positive returns in 2015.

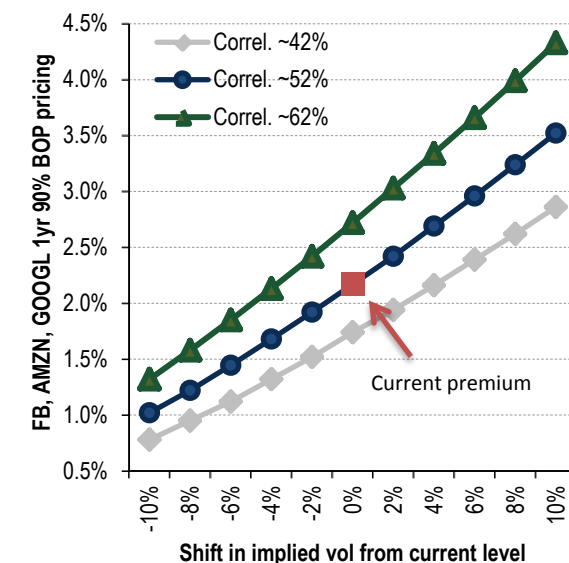
Lever divergence between relative price performance and implied vols of FANG vs. S&P500:

Despite the stellar outperformance of FANG relative to the S&P500 over the past three years, the average implied vol of these stocks is sitting at historically low levels vs. the index. This makes long vega structures such as best-of-puts attractive, all else being equal. For instance, Chart 54 shows that the ratio of the average 1yr ATM implied volatility of FB, AMZN, GOOGL to S&P is in its 27th percentile since May-12.

Buy implied correlation at a discount to realized: Chart 55 shows that the 1yr implied correl of FB, AMZN, and GOOGL is currently trading at a material discount to realized. At current levels of correlation, a 1yr 90% best of put on these three stocks costs 72% less than the average 1yr 90% vanilla put and 66% less than the least expensive put.

The trade is short delta, long vega, long correl: For instance, the mark-to-market of a best-of-put on FB, AMZN, GOOGL would gain 52bps if the implied vol of each of the stocks rises by 4% near-term, and 55bps if the implied correlation rises by 10pts (Chart 56).

Chart 56: The mark-to-market of a long best-of-put would benefit from a rise in implied volatility and correlation



Source: BofA Merrill Lynch Global Research. We assume parallel shift in vols / correls.

Low cost, long vol convexity hedge in VIX

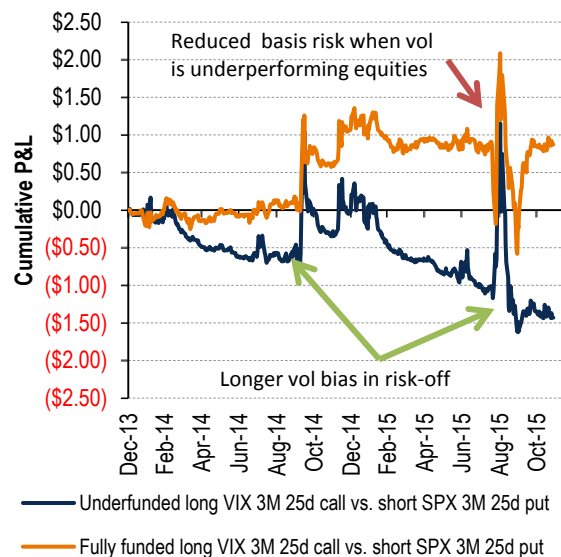
Trade: Buy VIX 3M 25d calls partially funded with short SPX 3M 25d puts.

Why? Systematically buying VIX OTM calls partially funded ('underfunded') with short SPX OTM puts would have provided more than 2x the hedge gains of the next best performing 'long VIX' strategy in Oct'14 and Aug-Sep'15 for a similar carry cost.

Underfunded vs. funded VIX/SPX hedge, longer volatility bias and reduced basis risk exposure

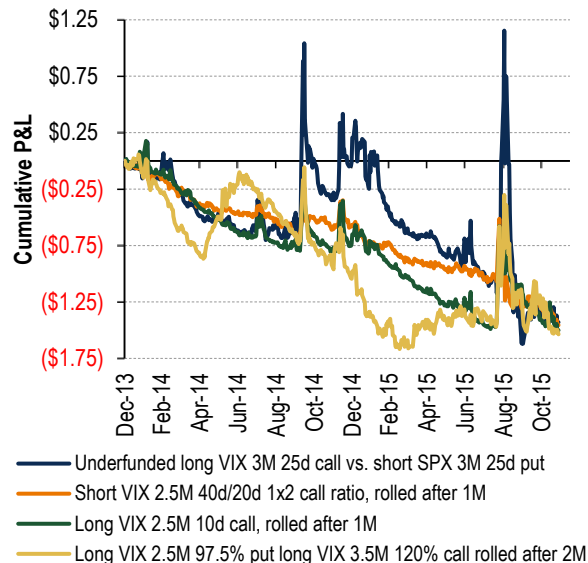
As described in our [vol convexity piece](#), fully funding long VIX OTM calls with short SPX OTM puts can create an efficient tail hedge with flat carry. The cost of this long convexity hedge is basis risk, the risk that volatility fails to rise sufficiently in an equity sell-off to compensate for losses on the short SPX put. The underfunded VIX/SPX strategy has a longer volatility exposure than its fully funded analog. Hence, it provides larger hedge benefits in risk-off events and is less likely to experience drawdowns when volatility is underperforming equities (Chart 57). The cost of this longer volatility bias and diminished exposure to basis risk is a larger cost of carry in quiet markets.

Chart 57: The underfunded VIX vs. SPX strategy would have provided larger hedge gains in equity sell-offs vs. its fully funded analog at the expense of a larger carry in quiet periods



Source: BofA Merrill Lynch Global Research. Daily data from 31-Dec-13 to 27-Nov-15. Indicative transaction costs are included.

Chart 58: The underfunded VIX vs. SPX strategy would have provided more than 2x the hedge gains of the next best performing popular 'long VIX hedge' in Oct'14 and Aug-Sep'15



Source: BofA Merrill Lynch Global Research. Daily data from 31-Dec-13 to 27-Nov-15. Strategies sized to have same carry cost since Dec-13. Indicative costs are included.

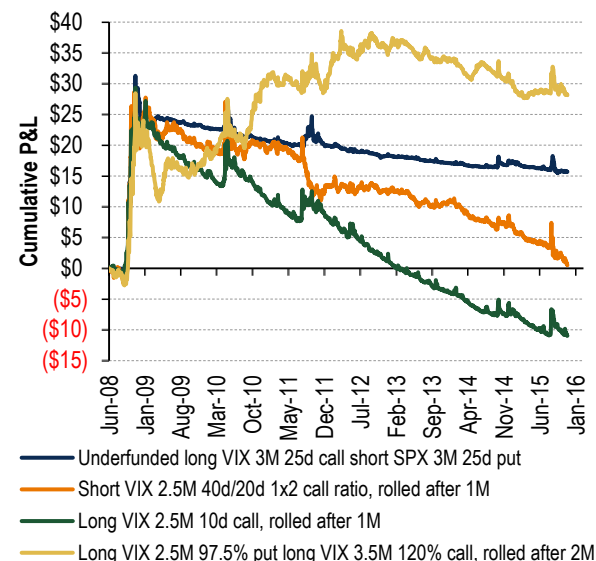
The VIX/SPX strategy generated larger hedge gains per unit of carry over recent periods

We size the following 'popular' systematic long volatility strategies to have the same cost of carry as the underfunded long VIX call vs. short SPX put strategy since Dec-13: (i) VIX 1by2 call ratio, (ii) VIX deep OTM call, and (ii) [VIX calendar strangle](#) (Chart 58). The underfunded VIX/SPX strategy would have been a superior hedge over the past 2 years generating materially larger gains across risk-off events for the same carry cost.

The VIX/SPX strategy achieved the best balance between hedge gains and carry historically

In Chart 59, we size the above strategies to provide similar hedge benefits in '08 and compare their long term performance. The underfunded VIX/SPX strategy would have provided (i) larger and more consistent hedge gains than the VIX 1by2 for a lower cost of carry, and (ii) a better balance between protection in risk-off events and decay in quiet periods than buying VIX deep OTM calls. The VIX calendar strangle would have outperformed from '08 to '12 but the performance profile of the strategy has shifted since 2013 with the put leg 'decaying' for extended periods of time. We see the main risk scenario for the underfunded VIX/SPX strategy as a prolonged grind lower in equities accompanied by a muted rise in volatility (e.g., Q4-07, H1-08).

Chart 59: The underfunded VIX vs. SPX strategy would have achieved a better balance historically between hedge gains in sell-offs and magnitude/consistency of decay in quiet markets



Source: BofA Merrill Lynch Global Research. Daily data from 17-Jun-08 to 27-Nov-15. Strategies sized to have similar hedge gains in '08. Indicative costs are included.

China upside and downside hedges

Upside hedge: Sell zero cost HSCEI Dec-16 1x2 ATM/114% call ratio (iv: 24.1/23.7)

Downside hedge 1: HSCEI Jun-16 95/75% put spread for 4.45% (4.5x payoff ratio)

Downside hedge 2: KOSPI2 Jun-16 90% put for 1.62% (iv: 18.6%)

Why? Further easing is expected and even QE is possible, creating upside risk to an under-owned market trading at low valuation. Meanwhile, downside is in demand on growth fears. Steep skew in HSCEI makes short call ratios cheap-to-carry upside hedges and put spreads efficient for downside. KOSPI2 is the best-value proxy hedge to HSCEI.

Chinese government does not want equities to go down, cannot rule out QE: With the rapid decline in growth, BofA-ML expects [Chinese policymakers to maintain an easing bias in monetary policy](#) to lower funding costs. Post the SDR inclusion, Beijing may slow its FX interventions and allow the CNY to fall more sharply. This could trigger competitive devaluations among EM countries, weigh on commodity prices and depress inflation break-evens. However, [this may open the door for the PBOC to implement its own version of QE at some stage](#) and could potentially bring value investors back into oversold EM assets, including the HSCEI.

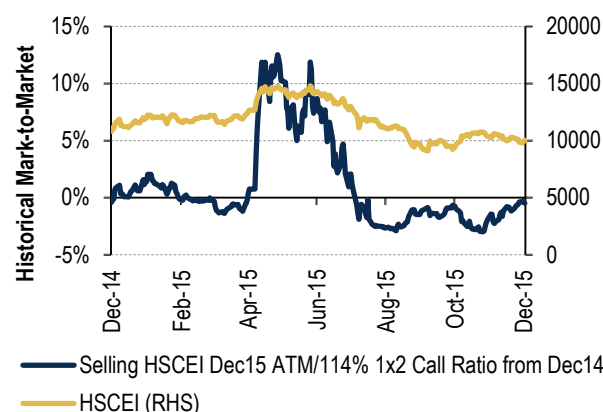
Steep HSCEI skew, upside volatility will get bid if the market rallies: HSCEI skew has steepened materially post the summer correction. Chart 60 shows that the current HSCEI 85-115% skew of 3.7% is at its 91st percentile since 2013. Structured products

Chart 60: HSCEI skew had a major spike in the summer of 2015 due to structured product issuers' buying



Source: BofA Merrill Lynch Global Research

Chart 61: Historical Mark-to-Market of selling HSCEI Dec15 ATM/114% 1x2 call ratio: Relatively low carry cost



Source: BofA Merrill Lynch Global Research

issuers have turned long skew as the HSCEI is now at the peak of the auto-callable outstanding vega. However, if the market rallies, the existing products will knock-out, creating a bid to volatility, while skew may get depressed by new issuance.

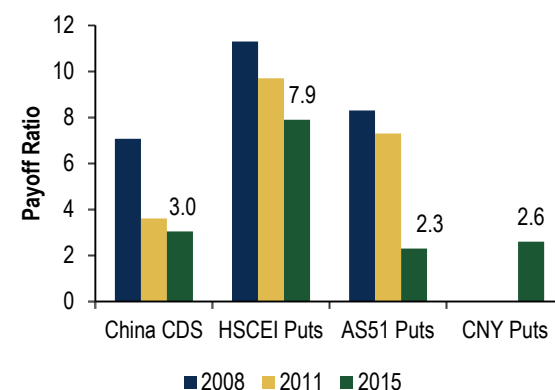
To hedge for a non-consensus upside scenario, we like selling HSCEI Dec-16 1x2 ATM/114% call ratio for zero cost (iv: 24.1/23.7). The trade is cheap to carry and offers an opportunity to participate in a potential HSCEI rally in 2016. Chart 61 shows the historical mark-to-market of a Dec-2015 expiry trade initiated in Dec-2014.

We see more downside risk in A-shares, but futures are the only liquid tool: China A-shares have more downside risk given a 40% premium vs. their H-shares counter-parts. However, with no liquid options market on China A-shares, FTSE A50 futures listed in Singapore are the most liquid tool with [a -10% annualized roll cost](#).

Hedge China downside with HSCEI Jun-16 95%/75% put spread: Chart 62 shows that HSCEI puts were the best hedges in previous crashes in 2008, 2011, 2015 compared to China CDS, AS51 puts or USDCNH calls. However, with HSCEI already trading below its 2008 P/B ratio, a recent 35% decline of the market, and the recent rise in volatility and skew, we prefer HSCEI put spreads to outright puts. Alternatively, we also like KOSPI2 puts given the cheap volatility and considering [it is the best proxy hedge for HSCEI](#).

FX hedge: BofA-ML has [high conviction on the RMB falling 9% in 2016](#) as China & US monetary policies diverge. The favorite FX trade is to go long USD/CNH 6M forward as the RMB may see a renewed drop as early as 1Q post the SDR inclusion & Fed hike.

Chart 62: Historical payoff of buying China hedges in previous market crashes



Source: BofA Merrill Lynch Global Research Assume buying 6-month 25delta put and 1-year China CDS at the peak and through during the 2008/2011/2015 crashes

KOSPI forward vol is cheapest globally

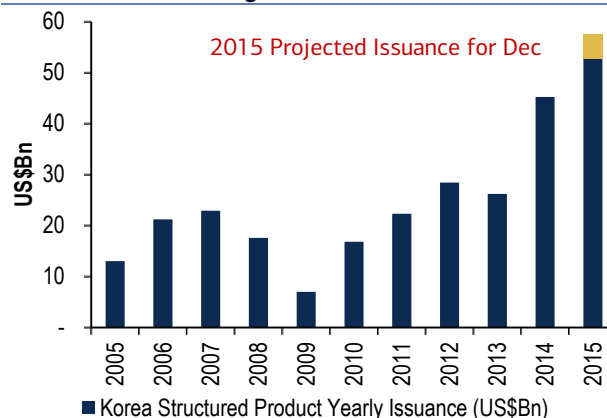
Trade: Buy KOSPI2 Jun-16/Jun-17 ATM forward vol agreement: 17.3% implied vol
Why? KOSPI2 vol has been the most subdued globally in the last 2 years owing to heavy structured product issuance; limited downside from current levels.

We also like forward vol in NKY as a Japan-specific Abenomics hedge, and in HSCEI but only at a lower entry level or vs SX5E.

Asian volatility has likely seen its lows in 2014-15 as lower structured product issuance in 2016 due to increasing regulation means less downward pressure going forward. Asian vol offers the best value given the flat term structure & low carry cost. KOSPI2 Jun-16/Jun-17 forward vol is trading at its 13th %-ile since 2008, the most subdued globally. NKY forward vol is an attractive hedge against an Abenomics failure. We also [continue to like](#) long HSCEI, short ESTX50 2y puts due to likely divergent vol dynamics.

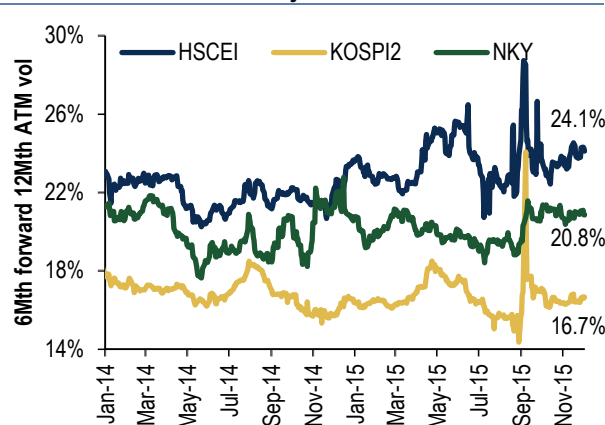
Asia is the best place to implement long volatility trades: With our view that volatility is on a gradual upward trend, we see buying Asian forward volatility as offering the best value. Asian volatility is particularly attractive as 1) Investors believe [China recession & EM debt crisis are the biggest global tail risks](#) in our Fund Manager Survey; 2) Asian term structures are the flattest globally due to volatility supply from auto-callables and hence, buying forward volatility is cheap to carry; 3) Asian markets have high downside beta to global markets and vol is more likely to spike in tail events as structured product issuers are short convexity embedded in the auto-callable structure

Chart 63: Korean non-principal protected structured product issuance hit an all-time high in 2015



Source: BofA Merrill Lynch Global Research

Chart 64: HSCEI long-dated forward volatility has reset higher vs. 2014 while KOSPI2 volatility offers the most value



Source: BofA Merrill Lynch Global Research

The 2016 long vol trade: Buy KOSPI2 Jun-16 / Jun-17 FVA: KOSPI2 volatility has been the most subdued globally in the last 2 years. KOSPI2 12M volatility starting in 6-months is trading in its 13th %-ile since 2008 or 24th %-ile since 2012, largely driven by the heavy structured product vega outstanding and the ultra-low intra-index correlation. By contrast, NKY/SPX/SX5E/HSCEI forward vols are all trading above their 60th %-iles since 2012. We prefer buying volatility through a KOSPI2 Jun16/ Jun17 ATM forward volatility agreement (FVA) at 17.3% implied volatility. We see limited downside from these levels but on the upside, investors can benefit from a gradual increase in global volatility or a HSCEI vol spike scenario similar to summer of 2015 (Chart 64).

We like owning HSCEI volatility but will target a lower entry level: While the HSCEI should have a higher market beta on the downside in a volatility spike scenario, we are worried about the potential downside risk in buying HSCEI volatility at current levels should China risk normalize. The HSCEI 6M-forward 12M ATM vol (24.1%) is currently in its 68th %-ile since 2012. We think there may be a better entry level in 1Q16 if the HSCEI stabilizes and auto-callables are re-issued. Currently, we prefer long HSCEI 2y 80% puts, short ESTX50 2y 80% puts as a hedge, both fundamentally (given China risk & ECB support) and technically (supportive vol dynamics from differences in outstanding vega profiles of HSCEI- & ESTX50-linked structured products).

Hedging a failure of Abenomics: Buy Nikkei Jun-16/Jun-17 forward volatility: BofA-ML's rates strategist expects JGB issuance in 2016 to decline YoY & the BOJ will purchase about 100% or more of issuance, up from 72% currently. [Policy sustainability will be questioned](#) after one last monetary and fiscal push in 1Q16. NKY Jun-16/Jun-17 ATM forward vol is an attractive hedge against an Abenomics failure scenario.

Table 5: KOSPI2 6-month forward 12-month ATM volatility is the lowest globally and is 2.5 vol points below the SPX

	6-Mth Forward 12-Mth ATM	%Tile Since 2012	%Tile Since 2008	12-Mth ATM
KOSPI2	16.7%	23.6%	12.3%	15.7%
NKY	20.8%	60.1%	35.8%	20.3%
SPX	19.3%	61.9%	32.2%	17.3%
SX5E	21.8%	62.0%	32.3%	21.4%
HSCEI	24.1%	68.2%	35.6%	24.8%

Source: BofA Merrill Lynch Global Research

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Potential Risk at Expiry & Options Limited Duration Risk

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<http://www.theocc.com/about/publications/character-risks.jsp>

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- When the underlying stock price meaningfully lies above the short call strike price, the investor is at risk of having their underlying shares called-away (sold). Holders can employ the following tactics to avoid call-away:

Roll-out (Roll Forward): Buy back the calls that were originally sold and sell same strike calls at a later expiry. Because of the time value, this transaction usually results in an initial net credit.

Roll-up: Buy back the calls that were originally sold and sell higher strike calls at the same expiry. Because of intrinsic value, this transaction usually results in a net debit.

Roll-up & out: Buy back the calls that were originally sold and sell higher strike calls at a later expiry. The transaction can result in either a net debit or credit.

- When the underlying stock price meaningfully underperforms, the investor will have an opportunity to shift strikes and/or expiries to generate further income.

Roll-out (Roll Forward): Buy back the calls that were originally sold and sell same strike calls at a later expiry. Because of the time value, this transaction usually results in an initial net credit.

Roll-down: Buy back the calls that were originally sold and sell lower strike calls at the same expiry. Because of intrinsic value, this transaction usually results in an initial net credit.

Roll-down & out: Buy back the calls that were originally sold and sell lower strike calls at a later expiry. Because of time value and intrinsic value, this transaction will result in an initial net credit.

General return risks associated with covered call strategies

While covered call strategies will underperform stocks in fast bull markets, they will still realize significant profits. Covered call strategies tend to outperform outright stock ownership in flat, down and slightly up markets.

Covered call strategies allow clients to increase their yield on equity positions in exchange for giving up some of the potential upside in the underlying securities. While covered call strategies will underperform stocks in fast bull markets, they will still realize significant profits. Covered call strategies tend to outperform outright stock ownership in flat, down and slightly up markets. Covered call strategies tend to dampen the effects of general market volatility by offsetting outlier returns. The payoff properties of the covered call, capped upside returns and reduced downside returns (offset by option premiums received), work to truncate holding period returns towards the mean and reduce the overall standard deviation.

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Investment rating	Total return expectation (within 12-month period of date of initial rating)	Ratings dispersion guidelines for coverage cluster*
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Neutral	≥ 0%	≤ 30%
Underperform	N/A	≥ 20%

* Ratings dispersions may vary from time to time where BofA Merrill Lynch Research believes it better reflects the investment prospects of stocks in a Coverage Cluster.

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